

[24/09/2026 10:31 AM] Imaanaa Harmee:

INTRODUCTION TO BIOCHEMISTRY

✱ Biochemistry is the branch of science that studies the **chemical composition of living organisms** and the **chemical reactions that occur inside living cells**.

© The maintenance of health depends on the harmonious regulation of biochemical reactions. Whenever these reactions become abnormal or disturbed, disease develops.

Therefore, biochemistry serves as the molecular foundation of medicine.

Definition

✱ Biochemistry is the study of:

- ✓ The chemical composition of living organisms.
- ✓ Biological molecules.
- ✓ Chemical reactions occurring inside living cells.
- ✓ Metabolic pathways that sustain life.

In simple words,

Biochemistry explains how life works at the molecular level. 

✱ Why is Biochemistry Important?

🔗 NOTE: Every activity performed by the human body depends on biochemical reactions.

Examples include:

♥ Contraction of the heart

Breathing

🧠 Transmission of nerve impulses

💪 Muscle contraction

🍲 Digestion of food

Blood clotting

🦴 Bone formation

🧬 DNA replication

💉 Hormone production

⚡ ATP production

Without these biochemical reactions, life cannot exist.

Importance of Biochemistry in Medicine

🌐 Biochemistry helps healthcare professionals to:

- ☒ Understand normal body function.
- ☒ Understand the causes of diseases.
- ☒ Diagnose metabolic disorders.
- ☒ Interpret laboratory investigations.
- ☒ Understand how drugs produce their effects.
- ☒ Monitor disease progression.
- ☒ Develop new treatment strategies.

Modern medicine is therefore built upon biochemical principles.

⚠️ *What Happens When Biochemical Reactions Become Abnormal?*

Disturbances in biochemical reactions lead to disease.

Some examples include:

Diabetes Mellitus

Insulin deficiency or insulin resistance prevents glucose from entering cells efficiently.

As blood glucose increases, hyperglycemia develops.

Long-standing hyperglycemia damages the kidneys, eyes, nerves, and blood vessels.

Hypercholesterolemia

Abnormal lipid metabolism increases cholesterol accumulation inside arteries.

This contributes to atherosclerosis, myocardial infarction, and stroke.

Phenylketonuria (PKU)

Deficiency of the enzyme phenylalanine hydroxylase causes toxic accumulation of phenylalanine.

Without treatment, severe brain damage may occur.

These clinical examples are added to improve understanding and are **not individually listed in the lecture**.

High-Yield Exam Pearl

Many diseases begin as **biochemical disorders long before structural abnormalities become visible**.

Example:

A diabetic patient may have elevated blood glucose for years before developing kidney disease or blindness.

Clinical Correlation

Many laboratory investigations routinely ordered in hospitals are biochemical tests.

Examples include:

 Blood glucose

Serum cholesterol

Cardiac enzymes

 Liver function tests

 Kidney function tests

 Electrolytes

These tests often detect disease before obvious clinical symptoms appear.

INTRODUCTION TO THE CELL

 Cell Is the **basic structural and functional unit of all living organisms.**

 Every living organism is composed of one or more cells.

 Cells perform all activities necessary for life.

 Because every tissue and every organ is composed of cells, the cell is considered the smallest unit capable of independent life.

Definition of a Cell

A cell is the smallest living unit capable of performing all life processes.

Every cell possesses:

 Genetic material

 Cytoplasm

 Plasma membrane

 Cellular machinery required for metabolism

Each cell is capable of maintaining its own internal environment while interacting with surrounding cells.

Why is the Cell Called the Structural Unit?

Every structure within the human body is ultimately made of cells.

Examples include:

 Cardiac muscle cells form the heart.

 Neurons and glial cells form the brain.

 Osteocytes form bone tissue.

[24/09/2026 10:31 AM] Imaanaa Harmee: Pneumocytes line the lungs.

Blood cells circulate within blood vessels.

Therefore,

Cells combine to form tissues.

Tissues combine to form organs.

Organs combine to form organ systems.

Organ systems together make up the human body.

Why is the Cell Called the Functional Unit?

Every physiological function depends on specialized cellular activity.

Examples include:

 Muscle cells produce movement.

🧠 Neurons conduct nerve impulses.

Red blood cells transport oxygen.

🛡️ White blood cells defend against infection.

💉 Endocrine cells secrete hormones.

🦴 Bone cells maintain skeletal strength.

If cells lose their normal function, organs cannot function normally.

🧬 Biological Organization

★ Living organisms are organized in a stepwise manner.

★ Small biological molecules combine to form organelles.

★ Organelles work together inside cells.

★ Groups of similar cells form tissues.

★ Different tissues combine to form organs.

★ Several organs cooperate to form organ systems.

★ Finally, all organ systems together form the complete human body.

🌱 Characteristics of Living Cells

Cells contain hereditary information that regulates cellular activities and allows transmission of information to daughter cells.

Living cells also possess several important characteristics.

They are able to:

☒ Grow

- ☒ Divide
- ☒ Produce energy
- ☒ Synthesize proteins
- ☒ Respond to environmental stimuli
- ☒ Remove waste products
- ☒ Maintain homeostasis
- ☒ Communicate with neighboring cells
- ☒ Repair damaged structures whenever possible

These characteristics distinguish living cells from non-living matter.

Size of Cells

-  Most animal and plant cells measure approximately **1–100 μm** .
-  Because of their small size, they can usually be observed only with a microscope.

Hereditary Information

Every cell contains hereditary information that regulates cellular activities.

Hereditary information is essential for regulating cell function and transmitting information to daughter cells during cell division.

Hereditary information determines:

-  Protein synthesis
-  Cell growth
-  Cell division

 Cellular differentiation

 Repair mechanisms

 Genetic inheritance

Hereditary information is stored in **DNA**, which is organized into genes and chromosomes.

✳ Genes provide the instructions needed to synthesize proteins, and proteins determine almost every structural and functional property of the cell.

Types of Cells

According to the lecture, living cells are divided into **two major groups**.

◇ Prokaryotic Cells

Prokaryotic cells are structurally simple cells that **lack a true nucleus**.

Their genetic material is located in a region called the **nucleoid** rather than within a membrane-bound nucleus.

Characteristics

- ☒ Small size
- ☒ Simple organization
- ☒ No membrane-bound organelles
- ☒ Circular DNA
- ☒ Binary fission for reproduction

Examples

 Bacteria

 Archaea

◇ Eukaryotic Cells

Eukaryotic cells are larger and structurally more complex.

Their DNA is enclosed within a membrane-bound nucleus.

They also contain numerous specialized organelles that perform specific cellular functions.

Characteristics

- ☒ Large size
- ☒ True nucleus
- ☒ Membrane-bound organelles
- ☒ Linear chromosomes
- ☒ Cell division by mitosis or meiosis

Examples

 Humans

 Plants

 Fungi

 Protozoa

Differences Between Prokaryotic and Eukaryotic Cells

Prokaryotic Cells

Lack a membrane-bound nucleus.

Lack membrane-bound organelles.

Possess circular DNA.

Have a relatively simple internal structure.

Reproduce by binary fission.

Eukaryotic Cells

Possess a membrane-bound nucleus.

Contain membrane-bound organelles.

Possess linear chromosomes.

Have a more complex internal organization.

Divide by mitosis or meiosis.

The lecture emphasizes that prokaryotic cells are **smaller and simpler**, whereas eukaryotic cells are **larger and more complex**.

Clinical Correlation

Many antibiotics work because bacteria are prokaryotic cells.

[24/09/2026 10:31 AM] Imaanaa Harmee:  They target bacterial structures that are absent in human cells,

allowing selective destruction of bacteria with relatively little damage to normal human cells.

Memory Trick

Remember the characteristics of living cells using
"GRAPES." 

G = Growth

R = Reproduction

A = ATP production

P = Protein synthesis

E = Environmental response

S = Signaling and communication

★ High-Yield Exam Pearls

★ Biochemistry studies the chemical basis of life.

★ Health depends on normal biochemical reactions.

★ Disturbance of biochemical reactions results in disease.

★ The cell is the structural and functional unit of life.

★ Most cells measure approximately **1–100 μm** .

★ Cells contain hereditary information that regulates cellular activities.

★ Living organisms are classified into **prokaryotic** and **eukaryotic** cells.

★ Prokaryotic cells lack a true nucleus and membrane-bound organelles.

★ Eukaryotic cells possess a true nucleus and specialized membrane-bound organelles.

★ Every cell is enclosed by an outer membrane that separates the intracellular environment from the extracellular environment.

[24/09/2026 10:31 AM] Imaanaa Harmee:

🕒 INTRODUCTION TO THE PLASMA MEMBRANE

Every living cell is surrounded by a thin, flexible membrane called the **plasma membrane** or **cell membrane**.

The plasma membrane acts as the **communication link between the cell and its environment**.

It separates the intracellular contents from the extracellular environment while allowing the cell to communicate with surrounding cells.

🌀 The plasma membrane is not simply a protective covering.

It is a **dynamic, living structure** that continuously regulates the movement of substances, receives signals from other cells, and helps maintain the normal internal environment of the cell.

📖 Definition of the Plasma Membrane

☑ The plasma membrane is a **thin, flexible, selectively permeable biological membrane** that surrounds every living cell.

It consists mainly of:

📖 Lipids

🧬 Proteins

🌀 Carbohydrates

These components work together to maintain normal cellular function.

✳️ Why is the Plasma Membrane Important?

The plasma membrane performs numerous essential functions.

It:

☑ Protects the cell from the external environment.

☑ Maintains the internal environment of the cell.

☑ Controls the movement of substances into and out of the cell.

☑ Receives chemical signals from hormones and neurotransmitters.

☑ Allows communication between neighboring cells.

☑ Helps cells recognize one another.

☑ Anchors the cytoskeleton and provides structural support.

☞ Without the plasma membrane, cells would lose their organization and would not survive.

🧬 **Chemical Composition of the Plasma Membrane**

biological membranes are composed mainly of:

📖 Phospholipids

 Sphingolipids

 Cholesterol

 Proteins

 Carbohydrates

Each component contributes specific structural and functional properties to the membrane.

Phospholipids

Phospholipids are the **major structural component** of biological membranes.

They form the basic framework of the plasma membrane.

Each phospholipid molecule contains two distinct regions.

Hydrophilic Head

The phosphate-containing head is **polar**.

Because it is attracted to water, it faces the aqueous environments both inside and outside the cell.

Hydrophobic Tails

The fatty acid tails are **nonpolar**.

They repel water and therefore face toward one another in the center of the membrane.

This arrangement produces the **phospholipid bilayer**, which forms the structural basis of all biological membranes.

Amphipathic Nature of Phospholipids

★ Phospholipids are described as **amphipathic molecules** because they possess both:

☒ A hydrophilic (water-loving) head.

☒ A hydrophobic (water-fearing) tail.

🔗 This property allows phospholipids to spontaneously organize into a bilayer whenever they are placed in water.

🧠 The Phospholipid Bilayer

Phospholipids arrange themselves into two layers.

The hydrophilic heads face the surrounding water.

The hydrophobic fatty acid tails face one another within the center of the membrane.

This arrangement creates a **hydrophobic barrier** that prevents many substances from crossing the membrane freely.

⚠️ *Why Can't All Molecules Cross the Membrane?*

The hydrophobic center of the membrane prevents many water-soluble substances from passing directly through it.

Small nonpolar molecules such as:

👉 Oxygen

👉 Carbon dioxide

can diffuse through the lipid bilayer easily.

In contrast, charged or polar molecules such as:

👉 Sodium

👉 Potassium

👉 Calcium

👉 Glucose

require specialized membrane proteins for transport.

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The **hydrophobic core of the phospholipid bilayer** is responsible for the selective

permeability of the plasma membrane.

Selective Permeability

The plasma membrane is **selectively permeable**, meaning it regulates which substances enter and leave the cell.

Selective permeability allows the cell to:

- ☒ Obtain nutrients.
- ☒ Remove waste products.
- ☒ Maintain ion balance.
- ☒ Preserve homeostasis.
- ☒ Protect itself from harmful substances.

Fluid Mosaic Model

The plasma membrane using the **Fluid Mosaic Model**.

[24/09/2026 10:31 AM] Imaanaa Harmee: This model explains both the structure and behavior of biological membranes.

Why is it Called "Fluid"?

The phospholipids and many membrane proteins are able to move laterally within the membrane.

This fluidity allows:

- ☒ Cell growth.
- ☒ Cell division.
- ☒ Membrane repair.
- ☒ Endocytosis.
- ☒ Exocytosis.

- ☒ Cell movement.

🧩 Why is it Called "Mosaic"?

The membrane contains many different proteins scattered throughout the phospholipid bilayer.

These proteins resemble pieces of a mosaic artwork.

Therefore, the membrane appears as a **fluid lipid sea with floating protein molecules**.

📦 Cholesterol

Cholesterol is present within biological membranes and **alters membrane fluidity** because of its rigid ring structure. In most membranes, it decreases membrane fluidity and contributes to membrane stability.

❄️ Functions of Cholesterol

- ☒ Stabilizes the membrane.
- ☒ Prevents excessive membrane movement.
- ☒ Reduces membrane permeability.
- ☒ Maintains membrane integrity.

🧬 Membrane Proteins

Proteins are responsible for most of the **functional activities** of the plasma membrane.

The lecture highlights several important functions of membrane proteins.

🌀 1. Enzymatic Activity

Some membrane proteins function as enzymes.

Example:

Na⁺/K⁺ ATPase

This enzyme helps maintain the normal sodium and potassium concentrations inside cells.

2. Carrier Function

Many substances cannot cross the lipid bilayer directly.

Carrier proteins transport molecules across the membrane.

Examples include

transport proteins called **translocases**.

3. Signal Transduction

Some membrane proteins function as receptors.

Hormones and other signaling molecules bind to these receptors.

This interaction initiates intracellular signaling pathways that alter cellular activity.

4. Structural Support

Some proteins anchor the plasma membrane to both the cytoskeleton and the extracellular matrix.

The lecture lists examples including:

 Fibronectin

 Ankyrin

 Spectrin

These proteins help maintain the structural integrity of cells and tissues.

5. Ion Channels

Certain membrane proteins form channels that regulate ion movement.

Examples include channels for:

-  Sodium
-  Potassium
-  Calcium
-  Chloride

These channels contribute to the **excitability of biological membranes**.

Membrane Carbohydrates

The lecture explains that membrane carbohydrates participate in:

- ☒ Cell recognition.
- ☒ Cell communication.
- ☒ Receptor functions.

Most membrane carbohydrates project outward from the cell surface, forming part of the **glycocalyx**, which contributes to cell recognition.

Integral Membrane Proteins

Integral proteins are firmly embedded within the phospholipid bilayer through hydrophobic interactions and van der Waals forces.

Their principal role is the transport of substances across the membrane.

Because they penetrate the membrane, these proteins are difficult to remove without disrupting the lipid bilayer.

Clinical Correlation

In **cystic fibrosis**, mutation of the CFTR chloride channel disrupts chloride transport across epithelial cell membranes.

This results in:

Thick, sticky mucus.

Recurrent respiratory infections.

Pancreatic dysfunction.

This illustrates how defects in membrane proteins can produce significant disease.

🧠 Memory Trick

Remember the four major membrane components with the mnemonic:

"People Prefer Chocolate Cookies." 🍪

People = Phospholipids

Prefer = Proteins

Chocolate = Cholesterol

Cookies = Carbohydrates

★ High-Yield Exam Pearls

★ The plasma membrane forms the boundary between the cell and its environment.

★ The plasma membrane is selectively permeable.

★ Phospholipids are amphipathic molecules.

★ The phospholipid bilayer forms the structural basis of biological membranes.

★ The hydrophobic core prevents the free passage of most polar molecules.

★ The Fluid Mosaic Model explains membrane structure.

[24/09/2026 10:31 AM] Imaanaa Harmee: ★ Cholesterol stabilizes the membrane and generally decreases membrane fluidity.

★ Membrane proteins function as enzymes, carriers, receptors, structural proteins, and ion channels.

★ Membrane carbohydrates are important for cell recognition and communication.

★ Integral membrane proteins are embedded within the phospholipid bilayer and mainly participate in transport.

[24/09/2026 10:31 AM] Imaanaa Harmee:

🧠 INTRODUCTION TO THE CYTOPLASM

The lecture explains that every cell contains a region called the **cytoplasm**, which occupies the space between the plasma membrane and the nucleus.

The cytoplasm contains numerous specialized structures called **organelles**, each performing specific functions necessary for cell survival.

The cytoplasm is not simply a fluid-filled space.

Instead, it is a highly organized environment where thousands of biochemical reactions occur every second.

Without the cytoplasm, cellular metabolism would not be possible.

📖 Definition of Cytoplasm

The **cytoplasm** is the portion of the cell located between the plasma membrane and the nucleus.

It consists of:

💧 Cytosol

🧬 Organelles

🧴 Inclusions such as stored nutrients and pigments

The cytoplasm provides the environment in which most cellular activities occur.

💧 Cytosol

The **cytosol** is the fluid component of the cytoplasm.

It contains:

☄ Water

✦ Dissolved proteins

✦ Electrolytes

✦ Sugars

✦ Amino acids

✦ Enzymes

✦ Metabolic intermediates

☒ Most metabolic reactions occurring outside organelles take place within the cytosol.

Examples include:

🔪 Glycolysis

🔪 Protein synthesis on free ribosomes (*Supplementary concept*)

🏢 Cell Organelles

The lecture introduces **cell organelles** as specialized intracellular structures that perform particular functions necessary for cell survival.

Each organelle functions like a department within a factory.

Together, they maintain the life of the cell.

🧬 Classification of Cell Organelles

Cell organelles can be divided into two major groups.

📦 *Membranous Organelles*

These organelles are surrounded by one or more membranes.

Examples include:

🧠 Nucleus

⚡ Mitochondria

🔬 Endoplasmic reticulum

📦 Golgi apparatus

♻️ Lysosomes

🔪 Peroxisomes

◊ **Non-Membranous Organelles**

These organelles are **not surrounded by membranes**.

Examples include:

🧬 Ribosomes

🦴 Cytoskeleton

🌀 Centrioles

🧠 **The Nucleus**

The nucleus is the **control center of the cell**.

It contains the hereditary material that regulates almost every cellular activity.

The lecture identifies the nucleus as one of the principal organelles of eukaryotic cells.

📖 Functions of the Nucleus

The nucleus performs several essential functions.

It:

☒ Stores DNA.

☒ Controls protein synthesis.

☒ Regulates cell division.

- ☒ Controls cellular metabolism.
- ☒ Determines cell differentiation.
- ☒ Regulates gene expression.

Without the nucleus, most cells cannot survive for long.

Components of the Nucleus

The nucleus consists of:

- ☒ Nuclear envelope
 - ☒ Nuclear pores
 - ☒ Chromatin
 - ☒ Nucleolus
-  These structures work together to regulate genetic activity.

★ High-Yield Exam Pearl

The nucleus contains **DNA**, which serves as the genetic blueprint of the cell.

Mitochondria

Mitochondria are commonly known as the **powerhouses of the cell**.

Their primary function is the production of ATP, which serves as the major energy currency of the body.

Cells requiring large amounts of energy contain numerous mitochondria.

Examples include:

-  Cardiac muscle cells
-  Skeletal muscle cells

Neurons

Functions of Mitochondria

Mitochondria are responsible for:

- ☒ ATP production.
- ☒ Aerobic respiration.
- ☒ Oxidative phosphorylation.
- ☒ Regulation of programmed cell death (apoptosis).

Clinical Correlation

Diseases affecting mitochondria often produce weakness in tissues with high energy requirements, including the brain and skeletal muscles.

Endoplasmic Reticulum (ER)

The endoplasmic reticulum is an interconnected network of membranous tubules extending throughout the cytoplasm.

It is divided into two major types.

◇ **Rough Endoplasmic Reticulum (RER)**

The rough endoplasmic reticulum contains numerous ribosomes attached to its surface.

Its primary function is **protein synthesis**.

Proteins synthesized by the RER are usually destined for:

Secretion.

Cell membranes.

Lysosomes.

◇ Smooth Endoplasmic Reticulum (SER)

The smooth endoplasmic reticulum lacks ribosomes.

Its major functions include:

☒ Lipid synthesis.

[24/09/2026 10:31 AM] Imaanaa Harmee: ☒ Steroid hormone synthesis.

☒ Detoxification of drugs.

☒ Calcium storage in muscle cells.

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Rough ER = Protein synthesis

Smooth ER = Lipid synthesis and detoxification

🧬 Ribosomes

Ribosomes are the **sites of protein synthesis**.

Unlike most organelles, ribosomes are **not surrounded by membranes**.

They may exist:

Free within the cytoplasm.

Attached to the rough endoplasmic reticulum.

Functions of Free Ribosomes

Free ribosomes synthesize proteins used within the cell itself.

Functions of Bound Ribosomes

Ribosomes attached to rough ER synthesize proteins that will be:

Secreted.

Inserted into membranes.

Transported to lysosomes.

Golgi Apparatus

The Golgi apparatus functions as the **processing and packaging center** of the cell.

Proteins synthesized by the rough ER are transported to the Golgi apparatus.

Within the Golgi apparatus they undergo:

- ☒ Modification.
- ☒ Packaging.
- ☒ Sorting.
- ☒ Distribution to their final destinations.

Functions of the Golgi Apparatus

The Golgi apparatus:

-  Modifies proteins.
-  Packages proteins.
-  Produces secretory vesicles.
-  Forms lysosomes.

Lysosomes

Lysosomes are membrane-bound organelles containing digestive enzymes.

These enzymes digest:

⚙️ Damaged organelles.

⚙️ Cellular debris.

⚙️ Foreign microorganisms.

🧴 Lysosomes therefore function as the **digestive system of the cell**.

Clinical Correlation

Deficiency of lysosomal enzymes produces **lysosomal storage diseases**, such as Tay–Sachs disease and Gaucher disease.

🔪 Peroxisomes

Peroxisomes contain enzymes responsible for oxidative reactions.

Their major functions include:

☒ Detoxification of harmful substances.

☒ Breakdown of hydrogen peroxide.

☒ Fatty acid metabolism.

One important enzyme is **catalase**, which converts hydrogen peroxide into water and oxygen.

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Remember:

Lysosomes digest.

Peroxisomes detoxify.

🧠 Memory Trick

Remember the major organelles using the mnemonic:

 "NMR GLP"

N = Nucleus

M = Mitochondria

R = Ribosomes

G = Golgi apparatus

L = Lysosomes

P = Peroxisomes

★ High-Yield Exam Pearls

★ Cytoplasm occupies the region between the plasma membrane and nucleus.

★ Cytosol is the fluid portion of the cytoplasm.

★ Organelles perform specialized cellular functions.

★ The nucleus controls cellular activities.

★ Mitochondria produce ATP.

★ Rough ER synthesizes proteins.

★ Smooth ER synthesizes lipids and participates in detoxification.

★ Ribosomes are the sites of protein synthesis.

★ The Golgi apparatus modifies, packages, and distributes proteins.

★ Lysosomes digest intracellular materials.

★ Peroxisomes detoxify harmful substances and contain catalase.

[24/09/2026 10:36 AM] Imaanaa Harmee:

 **INTRODUCTION TO CELL SIGNALING** 

Every cell in the human body must constantly communicate with other cells.

This communication process is called **cell signaling**.

Cell signaling allows cells to exchange information so that tissues and organs function together as one coordinated system.

Without cell signaling:

♥ The heart could not respond to adrenaline.

Blood glucose could not be regulated.

🧠 Nerve cells could not communicate.

🦴 Bones could not grow.

🛡️ The immune system could not recognize infection.

Therefore, **cell signaling is essential for life**.

📖 Definition of Cell Signaling

Cell signaling is the process by which one cell communicates with another through signaling molecules that bind to specific receptors and produce a cellular response.

Simply stated,

Cell signaling is the **language of cells**. 🧬

✿ Why is Cell Signaling Important?

Cell signaling regulates nearly every physiological process.

Examples include:

♥ Heart rate regulation

Blood pressure control

🧠 Glucose metabolism

🧠 Memory formation

💪 Muscle contraction

🦴 Bone growth

🛡️ Immune responses

🧬 Cell growth and differentiation

⚖️ Maintenance of homeostasis

💬 Components of Cell Signaling

Every signaling pathway contains several important components.

📁 Signaling Cell

The signaling cell produces and releases a signaling molecule.

Examples include:

🧠 Neurons

Endocrine cells

🛡️ Immune cells

✉️ Signaling Molecule (Ligand)

The signaling molecule is also called the **ligand**.

A ligand may be:

💉 Hormone

 Growth factor

 Neurotransmitter

 Cytokine

 Signaling molecules bind specifically to their appropriate receptors.

 Target Cell

The **target cell** is the cell that receives the signal.

A cell can respond only if it possesses the **appropriate receptor** capable of binding the ligand.

This explains why:

Insulin affects muscle and liver cells but not every cell equally.

Adrenaline acts strongly on the heart because cardiac cells contain β -adrenergic receptors.

 **Receptors**

A receptor is a specialized protein that recognizes and binds a specific signaling molecule.

receptors may be:

☒ Located inside the cell.

☒ Located on the cell surface.

Without receptors,

Cells cannot recognize signals.

 **The Three Steps of Cell Signaling**

Every signaling pathway consists of three major steps.

🔑 Step One: Signal Detection and Receptor Binding

A signaling cell secretes a signaling molecule into the extracellular space.

The ligand travels until it encounters a target cell possessing the correct receptor.

The receptor specifically recognizes and binds the ligand.

Only target cells possessing the correct receptor respond.

This step determines the **specificity** of cell signaling.

🔑 Step Two: Signal Transduction

After ligand binding,

The receptor changes its shape or activity.

This activates **intracellular signaling pathways**, also called **signal transduction cascades**, which amplify the original signal by producing multiple

intracellular signals.

🔑 What is Signal Amplification?

One extracellular signal can generate thousands of intracellular signaling molecules.

⚙️ Therefore,

A very small amount of hormone can produce a very large biological response.

This phenomenon is known as **signal amplification**, which explains the high sensitivity of hormonal signaling.

🔑 Step Three: Cellular Response

The signal transduction pathway finally produces a **specific cellular response**.

Examples include:

🧬 Altered gene expression.

⚙️ Activation or inhibition of enzymes.

🔑 Initiation of biochemical reactions inside the cell.

★ High-Yield Exam Pearl

Every signaling pathway follows the same general sequence:

🔑 Signal recognition.

🔑 Signal transduction.

🔑 Cellular response.

🧬 Types of Receptors

There are two broad receptor locations.

◇ Intracellular Receptors

These receptors are found:

Inside the cytoplasm.

Inside the nucleus.

They bind **small hydrophobic molecules** that easily cross the plasma membrane.

✳️ Molecules Using Intracellular Receptors

🔑 Cortisol

🔑 Steroid sex hormones

🔑 Vitamin D

[24/09/2026 10:36 AM] Imaanaa Harmee:  Thyroid hormone

 Retinoids

 Nitric oxide

Because these molecules are lipid-soluble,

They diffuse through the plasma membrane without requiring membrane transport proteins.

Clinical Correlation

Steroid hormones circulate in blood attached to carrier proteins because they are poorly soluble in water.

After entering target cells,

They bind intracellular receptors and regulate gene transcription.

Cell Surface Receptors

Most receptors are located on the cell membrane.

These receptors bind signaling molecules that cannot cross the lipid bilayer.

Examples include:

◆ Insulin

◆ Glucagon

◆ Growth factors

◆ Most peptide hormones

Mediators of Cell Surface Signaling

Signaling through cell surface receptors involves several important mediators.

They include:

 Ligands

 Receptors

 Transducers

 Effector molecules

 Second messengers

 Final target molecules

Each component contributes to successful transmission of the signal.

Ligand

The ligand initiates signaling.

Examples include:

Hormones

Growth factors

Receptor

The receptor specifically recognizes the ligand.

Examples

GPCR

RTK

Transducers

Transducers carry the signal from the receptor toward intracellular effectors.

Examples:

G proteins

Ras proteins

Effector Molecules

Effector proteins generate intracellular signals.

Examples:

Adenylate cyclase

MAP kinase (MAPK)

Second Messengers

Second messengers spread the signal rapidly throughout the cell.

The lecture lists:

cAMP

cGMP

Calcium ions (Ca^{2+})

Final Target Molecules

Ultimately,

Second messengers regulate final target molecules such as:

DNA

Ion channels

Enzymes

These targets produce the final cellular response.

Major Classes of Cell Surface Receptors

There are **three major classes of cell surface receptors**.

1 Protein-Coupled Receptors (GPCRs)

These are the **largest family** of cell surface receptors.

Characteristics include:

- ☒ Linked to trimeric G proteins.
- ☒ Ligand binding activates G proteins.
- ☒ Can activate enzymes.
- ☒ Can open ion channels.

2 Enzyme-Linked Receptors

Characteristics include:

- ☒ Possess enzyme activity or are linked to enzymes.
- ☒ Most commonly associated with protein kinases.
- ☒ Produce intracellular phosphorylation reactions.

3 Ion Channel-Linked Receptors

Characteristics include:

- ☒ Open ion channels after ligand binding.
- ☒ Especially important during synaptic transmission.
- ☒ Neurotransmitters temporarily open or close the channels.

 Memory Trick

Remember the three receptor classes using:

☞ "GEI"

G = G protein-coupled receptors

E = Enzyme-linked receptors

I = Ion channel-linked receptors

★ High-Yield Exam Pearls

★ Cell signaling enables communication between cells.

★ Every signaling pathway has three major steps.

★ Ligands bind only to specific receptors.

★ Target cells must possess appropriate receptors.

★ Signal transduction amplifies the original signal.

★ Cellular responses include enzyme activation and altered gene expression.

★ Receptors are either intracellular or located on the cell surface.

★ Intracellular receptors bind small hydrophobic molecules such as steroid hormones and thyroid hormone.

★ Cell surface signaling involves ligands, receptors, transducers, effectors, second messengers, and target molecules.

★ The three major classes of cell surface receptors are GPCRs, enzyme-linked receptors, and ion channel-linked receptors.

[24/09/2026 10:36 AM] Imaanaa Harmee:

📖 **INTRODUCTION TO G PROTEIN-COUPLED RECEPTORS (GPCRs)**

G protein-linked receptors (GPCRs) are the largest family of cell surface receptors.

They respond to many different extracellular signals and convert these signals into intracellular responses through G proteins.

✱ Characteristics of GPCRs

The lecture states that GPCRs have the following characteristics:

- ☒ They are very abundant.
- ☒ They consist of **seven membrane-spanning alpha helices** (serpentine receptors).
- ☒ They are linked to **trimeric G proteins**.
- ☒ They mediate responses to many different signals.
- ☒ They increase the concentration of second messengers such as **cAMP** by activating adenylate cyclase.

🧬 Structure of GPCRs

each GPCR:

Passes through the plasma membrane **seven times**.

Has an extracellular region that binds the ligand.

Has an intracellular region that interacts with G proteins.

Because of their seven membrane-spanning regions, GPCRs are also called **serpentine receptors**.

🧬 Structure of G Proteins

The lecture explains that G proteins consist of **three protein subunits**.

These are:

Alpha (α)

⊗ Beta (β)

Gamma (γ)

Together they form a **heterotrimeric G protein**.

⚙️ Inactive State of the G Protein

Before receptor stimulation,

the **alpha subunit binds GDP**.

🔑 In this condition,

the three subunits remain together,
and the G protein is **inactive**.

🚀 Activation of the G Protein

After a ligand binds to the GPCR,
the receptor activates the G protein.

The alpha subunit releases GDP.

GTP binds in place of GDP.

The alpha subunit separates from the beta-gamma complex.

Both the alpha subunit and the beta-gamma complex become active signaling components.

🎯 Function of Active Alpha Subunit

The **α -GTP complex** strongly stimulates **adenylate cyclase**.

Adenylate cyclase then produces the second messenger **cAMP**.

🔑 Adenylate Cyclase

Adenylate cyclase is a membrane-associated enzyme.

its function is to synthesize **cyclic AMP (cAMP)** from ATP.

🗨️ cAMP (Cyclic Adenosine Monophosphate)

cAMP is as an important **second messenger**.

After being produced,

cAMP activates **Protein Kinase A (PKA)**, allowing the signal to spread throughout the cell.

⚡ Protein Kinase A (PKA)

PKA contains:

- ☒ Two regulatory subunits.
- ☒ Two catalytic subunits.

🔒 Activation of Protein Kinase A

Two molecules of cAMP bind to each regulatory subunit.

This produces conformational changes.

The catalytic subunits are released.

The catalytic subunits become enzymatically active.

🌀 Functions of Active PKA

active catalytic subunits phosphorylate **serine** and **threonine** residues on target proteins located within the cytosol or nucleus.

🧬 PKA and Gene Expression

The active catalytic subunits enter the nucleus.

Inside the nucleus,

they phosphorylate the transcription factor **CREB**.

Activated CREB stimulates transcription of **cAMP-inducible genes**.

🔑 CREB

CREB stands for

cAMP Response Element-Binding Protein.

CREB activates transcription after phosphorylation by PKA.

🔗 PKA and Glycogen Metabolism

Example

adrenaline signaling in skeletal muscle.

Adrenaline binds β -adrenergic receptors.

This increases cAMP.

PKA becomes activated.

PKA phosphorylates several target proteins.

These events maximize glucose availability for ATP production.

● Termination of GPCR Signaling

1 GTP Hydrolysis

The alpha subunit possesses intrinsic **GTPase activity**.

It hydrolyzes GTP to GDP.

The inactive alpha subunit reassociates with the beta-gamma complex.

2 Protein Phosphatase 1

Protein Phosphatase 1 removes phosphate groups added by PKA.

This reverses phosphorylation and terminates the cellular response.

3 Breakdown of cAMP

cAMP is degraded by **phosphodiesterases**.

This decreases intracellular cAMP concentration and terminates signaling.

● Phosphodiesterase Inhibitors

phosphodiesterases are inhibited by:



Caffeine



Theophylline



Aminophylline

These drugs increase the effect of cAMP because they prevent its degradation.



Stimulatory and Inhibitory G Proteins

adenylate cyclase may be:



Activated by stimulatory G proteins.

[24/09/2026 10:36 AM] Imaanaa Harmee: ☒ Inhibited by inhibitory G proteins (Gi).



Memory Trick

Remember the GPCR pathway using:

"R-G-A-cAMP-PKA."

R = Receptor

G = G protein

A = Adenylate cyclase

cAMP = Second messenger

PKA = Protein Kinase A



High-Yield Exam Pearls



GPCRs are the largest family of cell surface receptors.



GPCRs possess seven transmembrane alpha helices.



G proteins contain alpha, beta, and gamma subunits.



GDP-bound alpha subunits are inactive.



GTP binding activates the alpha subunit.



Active alpha stimulates adenylate cyclase.

- ★ Adenylate cyclase synthesizes cAMP.
- ★ cAMP activates Protein Kinase A.
- ★ PKA phosphorylates serine and threonine residues.
- ★ CREB regulates transcription of cAMP-inducible genes.
- ★ Phosphodiesterases degrade cAMP.
- ★ Protein phosphatase 1 reverses phosphorylation.
- ★ Gi inhibits adenylate cyclase.

[24/09/2026 10:36 AM] Imaanaa Harmee:

📖 INTRODUCTION TO ENZYME-LINKED RECEPTORS 🧬

Enzyme-linked receptors are one of the three major classes of cell surface receptors.

Unlike GPCRs, these receptors possess **intrinsic enzyme activity** or are closely associated with intracellular enzymes.

After ligand binding, they initiate intracellular signaling by activating protein kinases.

✿ Characteristics of Enzyme-Linked Receptors

Enzyme-linked receptors ARE receptors that:

- ☒ Span the plasma membrane once.
- ☒ Bind extracellular signaling molecules.
- ☒ Activate intracellular enzyme activity.
- ☒ Commonly function through **protein kinases**.
- ☒ Regulate cell growth, metabolism, differentiation, and survival.

📖 Receptor Tyrosine Kinases (RTKs)

The **most common enzyme-linked receptors** are receptor tyrosine kinases (RTKs).

These receptors become activated after ligand binding and phosphorylate specific **tyrosine residues** on proteins.

Basic Structure of RTKs

Each receptor contains:

- An extracellular ligand-binding domain.

- A single membrane-spanning region.

-  An intracellular tyrosine kinase domain.

Each part contributes to signal transmission from outside the cell to the interior.

Ligand Binding and Receptor Dimerization

Ligand binding brings **two receptor molecules together**, producing **receptor dimerization**.

Dimerization is an essential step because it activates the intracellular kinase domains.

Autophosphorylation

Following dimerization, the intracellular kinase domains phosphorylate one another on **tyrosine residues**.

This is referred as **autophosphorylation**.

Autophosphorylation creates docking sites for intracellular signaling proteins.

Signal Transduction

The phosphorylated receptors recruit intracellular signaling proteins, initiating signaling pathways that regulate cellular activity.

These pathways transmit the signal from the membrane to intracellular targets.

Cellular Responses

Activation of RTKs produces important biological effects, including:

- ☒ Regulation of gene expression.

- ☑ Cell growth.
- ☑ Cell differentiation.
- ☑ Cellular metabolism.
- ☑ Cell survival.
- ✍ Insulin Receptor

The **insulin receptor** is an important example of an enzyme-linked receptor.

Binding of insulin activates the receptor's intracellular kinase activity and initiates downstream signaling.

🔑 Insulin Signaling

insulin receptor activation results in intracellular signaling that regulates metabolic activities.

The lecture emphasizes the receptor-mediated mechanism rather than passive diffusion.

🧠 Memory Trick

Remember RTK activation with:

🔑 **"Bind → Pair → Phosphorylate → Respond."**

Bind = Ligand binds receptor.

Pair = Receptor dimerization.

Phosphorylate = Tyrosine autophosphorylation.

Respond = Cellular response.

★ High-Yield Exam Pearls

- ★ Enzyme-linked receptors are one of the three major classes of cell surface receptors.
- ★ Receptor tyrosine kinases are the most common enzyme-linked receptors.

★ RTKs contain extracellular, transmembrane, and intracellular kinase domains.

★ Ligand binding induces receptor dimerization.

★ Dimerization leads to autophosphorylation of tyrosine residues.

★ Autophosphorylation activates intracellular signaling pathways.

★ The insulin receptor is an important example of a receptor tyrosine kinase.

[24/09/2026 10:36 AM] Imaanaa Harmee:

📖 INTRODUCTION TO ION CHANNEL-LINKED RECEPTORS ⚡

Ion channel-linked receptors represent the **third major class of cell surface receptors**.

They are also called **ligand-gated ion channels (LGICs)** or **ionotropic receptors**.

Among all receptor-mediated signaling pathways, they are described as the **simplest and fastest** because ligand binding directly controls the opening or closing of an ion channel.

🌟 Why Are Ion Channel-Linked Receptors the Fastest?

Unlike GPCRs and enzyme-linked receptors, ion channel-linked receptors **do not require multiple intracellular signaling intermediates** before producing a response.

Instead,

A chemical messenger binds directly to the receptor.

The receptor immediately changes its shape.

The ion channel opens or closes.

Ions rapidly move across the membrane.

The membrane potential changes within milliseconds.

This direct mechanism allows **very rapid communication** between excitable cells.

📖 Definition of Ligand-Gated Ion Channels (LGICs)

The lecture defines ligand-gated ion channels as **transmembrane ion-channel proteins** that open in response to the binding of a chemical messenger such as a neurotransmitter.

Unlike voltage-gated channels, these channels are activated **by ligand binding rather than changes in membrane potential**.

The comparison with voltage-gated channels is added for understanding and is not explicitly discussed in the lecture.

Structure of Ligand-Gated Ion Channels

LGICs are **transmembrane proteins** that span the plasma membrane and form a channel through which ions can pass after ligand binding.

These receptors possess:

- An extracellular ligand-binding site.

-  A membrane-spanning ion channel.

-  A gating mechanism that opens or closes the channel after ligand binding.

Mechanism of Action

Step 1: Ligand Binding

A chemical messenger, usually a **neurotransmitter**, binds to the extracellular portion of the receptor.

Step 2: Channel Opening

Ligand binding produces a conformational (shape) change in the receptor.

As a result,

the ion channel opens.

Step 3: Ion Movement

Once opened,
specific ions move across the plasma membrane through the channel.

These ions include:

- 🧪 Sodium (Na^+)
- 🧪 Potassium (K^+)
- 🧪 Calcium (Ca^{2+})
- 🧪 Chloride (Cl^-)

🧩 Step 4: Cellular Response

Movement of ions changes the electrical properties of the cell membrane.

This produces a rapid physiological response, particularly in electrically excitable cells such as neurons and muscle cells.

★ High-Yield Exam Pearl

Ion channel-linked receptors produce **the fastest cellular responses** because the receptor itself functions as an ion channel.

⚡ Major Ions Passing Through Ligand-Gated Channels

Ligand-gated ion channels may allow the passage of:

- 🧪 Sodium ions (Na^+)
- 🧪 Potassium ions (K^+)
- 🧪 Calcium ions (Ca^{2+})
- 🧪 Chloride ions (Cl^-)

Each channel is selective for one or more of these ions.

🧠 Synaptic Signaling

Ion channel-linked receptors play an essential role in **synaptic signaling between electrically excitable cells**.

A neuron transmits an electrical impulse along its axon.

When the impulse reaches the synapse,

a neurotransmitter is released.

The neurotransmitter binds to ligand-gated ion channels on the target cell.

The ion channel opens.

Ion movement produces a rapid electrical response.

Example: Neuromuscular Junction

muscle cells respond to signals coming from a neighboring nerve cell.

The sequence described is:

 A nerve impulse reaches the nerve terminal.

 The nerve cell releases a neurotransmitter into the **synaptic cleft**.

 The neurotransmitter binds receptors on the muscle cell.

 Ligand-gated ion channels open.

 The muscle cell receives the signal and responds.

Synaptic Cleft

The **synaptic cleft** is the space between the nerve cell and the muscle cell into which neurotransmitters are released.

This space allows chemical communication between two cells without direct physical continuity.

Physiological Importance

Ion channel-linked receptors are essential for:

☒ Rapid communication between nerve cells.

[24/09/2026 10:36 AM] Imaanaa Harmee: ☒ Signal transmission at neuromuscular junctions.

- ☑ Fast responses in electrically excitable tissues.
- ☑ Conversion of chemical signals into electrical signals.

🧠 Memory Trick

Remember the sequence using:

⚡ "**Bind → Open → Flow → Respond.**"

Bind = Neurotransmitter binds receptor.

Open = Ion channel opens.

Flow = Ions move across the membrane.

Respond = Electrical signal is generated.

⚠ Common Exam Traps

✗ **Trap 1:** Thinking ion channel-linked receptors require second messengers.

✓ **Correction:** The lecture presents them as the simplest and fastest signaling pathway because ligand binding directly opens the channel.

✗ **Trap 2:** Thinking all receptors activate protein kinases.

✓ **Correction:** Ion channel-linked receptors primarily regulate ion flow rather than activating kinase cascades.

✗ **Trap 3:** Thinking only sodium passes through ligand-gated channels.

✓ **Correction:** The lecture lists **Na⁺, K⁺, Ca²⁺, and Cl⁻** as ions that may pass through ligand-gated ion channels.

? Possible Viva Questions

? What are ligand-gated ion channels?

- ? Why are ion channel-linked receptors considered the fastest signaling pathway?
- ? Which ions commonly pass through ligand-gated ion channels?
- ? What is the role of ligand-gated ion channels in synaptic transmission?
- ? What is the synaptic cleft?
- ? Explain signaling at the neuromuscular junction.

★ High-Yield Exam Pearls

★ Ion channel-linked receptors are also called **ligand-gated ion channels (LGICs)** or **ionotropic receptors**.

★ They are the **simplest and fastest** signaling pathway.

★ Ligand binding directly opens the ion channel.

★ Common ions transported include **Na⁺, K⁺, Ca²⁺, and Cl⁻**.

★ These receptors are essential for **synaptic signaling**.

★ Neurotransmitters act as the ligands.

★ The neuromuscular junction is a classic example discussed in the lecture.

★ The **synaptic cleft** is the space between the nerve cell and the muscle cell where neurotransmitters are released.

[24/09/2026 10:36 AM] Imaanaa Harmee: 📖 **Carbohydrates**

🔗 Definition

Carbohydrates are organic compounds composed primarily of carbon (C), hydrogen (H), and oxygen (O).

They are **polyhydroxy aldehydes or ketones**, or compounds that yield these molecules upon hydrolysis

Carbohydrates are polyhydroxy aldehydes or ketones, or substances that produce them on hydrolysis.

Why are they called Saccharides?

The word "**Saccharide**" comes from the Greek word "**Sakkharon (σάκχαρον)**", meaning "**Sugar.**"

Examples:

🔗 Monosaccharide = One sugar unit

- 🔗 Disaccharide = Two sugar units
- 🔗 Oligosaccharide = Few sugar units
- 🔗 Polysaccharide = Many sugar units

Composition

Carbohydrates mainly contain:

- Carbon (C)
- Hydrogen (H)
- Oxygen (O)

Most carbohydrates follow the empirical formula:



where $n \geq 3$

Examples

Carbohydrate Molecular Formula

Glyceraldehyde $C_3H_6O_3$

Ribose $C_5H_{10}O_5$

Glucose $C_6H_{12}O_6$

Fructose $C_6H_{12}O_6$

⚠ Exam Trap

Many students think **all carbohydrates follow $(CH_2O)_n$** .

🚫 This is **not always true**.

Examples:

- ✓ **Deoxyribose ($C_5H_{10}O_4$)** ✗ Does **not** fit the formula.
- ✓ Amino sugars (e.g., glucosamine) contain **nitrogen (N)**.
- ✓ Sugar phosphates (e.g., glucose-6-phosphate) contain **phosphorus (P)**.
- ✓ Sulfated glycosaminoglycans contain **sulfur (S)**.

💡 Therefore, carbohydrates may also contain:

- Nitrogen (N)
- Phosphorus (P)
- Sulfur (S)

💡 **High Yield:** Do not define carbohydrates solely by the empirical formula; the structural definition (polyhydroxy aldehydes or ketones) is more accurate.

Why Are Hydroxyl (-OH) Groups Important?

▽ Carbohydrates contain many **hydroxyl (-OH)** groups.

These groups make carbohydrates:

- 💧 Highly soluble in water
- 🤝 Capable of forming hydrogen bonds
 - Easily transported in blood (e.g., glucose)
- 🧬 Able to interact with enzymes and receptors

🌾 Primary Source of Carbohydrates

🌱 Plants are the primary source.

Plants synthesize glucose by **photosynthesis**.

🌱 Storage in Plants

Plants store excess glucose as:

🌱 **Starch**

Examples:

🌱 🌽 Corn

🌱 🍚 Rice

🌱 🌾 Wheat

🌱 🥔 Potato

These are major dietary sources of carbohydrates for humans.

Carbohydrates in Animals

✓ Animals do **not** store glucose as starch.

✓ Instead, excess glucose is stored as:

🌱 **Glycogen**

📍 Liver → Maintains blood glucose between meals.

📍 Skeletal muscle → Provides energy for muscle contraction.

Plants 🌱 Store glucose as **Starch**

Animals 🌱 Store glucose as **Glycogen**

🧬 Monosaccharides

📖 Definition

A **monosaccharide** is the simplest carbohydrate that **cannot be hydrolyzed into smaller carbohydrate units**.

They are **polyhydroxy aldehydes or ketones**.

Examples:

🌱 Glucose

🌱 Fructose

🌱 Galactose

🌱 Ribose

🌱 Deoxyribose

🔍 Why Are They Called Aldehydes or Ketones?

Monosaccharides are classified based on their carbonyl group:

Aldoses

Contain an **aldehyde (-CHO)** group.

Examples:

- ◆ Glucose
- ◆ Galactose
- ◆ Ribose

Ketoses

Contain a **ketone (>C=O)** group.

Examples:

- ◆ Fructose
- ◆ Dihydroxyacetone

[24/09/2026 10:36 AM] Imaanaa Harmee:

🌀 Exam Piont

- 👉 **Most abundant monosaccharide in the human body: Glucose**
- 👉 **Main storage carbohydrate in plants: Starch**
- 👉 **Main storage carbohydrate in animals: Glycogen**
- 👉 **Only energy source for RBCs: Glucose (via anaerobic glycolysis)**
- 👉 **Energy yield of carbohydrates: 4 kcal/g**

🌀 Not all carbohydrates strictly follow the empirical formula **(CH₂O)_n**; use the structural definition for accuracy.

[24/09/2026 10:36 AM] Imaanaa Harmee:

🧬 Aldoses vs Ketoses

🌀 What is the Carbonyl Group?

The **carbonyl group (C=O)** is the functional group that determines whether a monosaccharide is an **aldose** or a **ketose**.

There are two types:

Aldehyde Group

Structure:

-CHO

- ▽ Located at the **end** of the carbon chain.
- ▽ Makes the sugar an **aldose**.

Ketone Group

Structure:

>C=O

- ▽ Located **within** the carbon chain (usually at **C-2**).
- ▽ Makes the sugar a **ketose**.

Aldoses

An **aldose** is a monosaccharide containing an **aldehyde (-CHO)** group.

📍 In a **Fischer projection**, the aldehyde carbon is assigned **Carbon 1 (C-1)**.

💧 **Notice**

The **most oxidized carbon** is always numbered first.

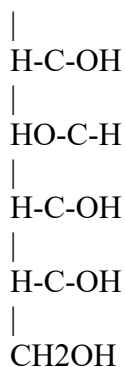
For aldoses:

➡ **Most oxidized carbon = C-1**

Example

🧬 Glucose

CHO ← C-1 (Aldehyde)



☑ Carbonyl carbon = C-1

Therefore, **Glucose is an Aldohexose**.

Other Examples

🧬 Glucose

🧬 Galactose

🧬 Ribose

🧬 Glyceraldehyde

Ketoses

A **ketose** is a monosaccharide containing a **ketone (>C=O)** group.

In most naturally occurring ketoses:

📍 The ketone is located on **Carbon 2 (C-2)**.

💧 **Notice**

The **most oxidized carbon** is **usually C-2** in ketoses because the carbonyl group is internal.

Example

🌀 Fructose

CH₂OH

|

C=O ← C-2 (Ketone)

|

HO-C-H

|

H-C-OH

|

H-C-OH

|

CH₂OH

☑ Carbonyl carbon = **C-2**

Therefore, **Fructose is a Ketohexose.**

Other Examples

🌀 Fructose

🌀 Dihydroxyacetone

🌀 Ribulose

🌀 Xylulose

[24/09/2026 10:36 AM] Imaanaa Harmee:

Asymmetric (Chiral) Carbon Atoms

A **chiral (asymmetric) carbon** is a carbon atom bonded to **four different groups**.

🌀 Chiral carbons give rise to **stereoisomers**, which have the same molecular formula but different three-dimensional arrangements.

🌀 Why Do Ketoses Have One Fewer Chiral Center?

The carbonyl carbon itself is **not chiral** because it is double-bonded to oxygen.

Aldoses

The aldehyde group is at **C-1**, so the remaining internal carbons (except the terminal CH₂OH carbon) can be chiral.

For an **aldohexose** (e.g., glucose):

C-1 = carbonyl (not chiral)

C-2, C-3, C-4, C-5 = chiral

C-6 = CH₂OH (not chiral)

➡ **Total chiral centers = 4**

Ketoses

The ketone group is at **C-2**, so one additional carbon cannot be chiral.

For a **ketohexose** (e.g., fructose):

C-1 = CH₂OH (not chiral)

C-2 = carbonyl (not chiral)
C-3, C-4, C-5 = chiral
C-6 = CH₂OH (not chiral)
➡ **Total chiral centers = 3**

💧 Notice

For monosaccharides with the same number of carbons:

Ketoses have one fewer chiral center than aldoses.

[24/09/2026 10:40 AM] Imaanaa Harmee:

🔗 Stereoisomerism of Monosaccharides

Stereoisomerism is the phenomenon in which compounds have:

- ✓ The **same molecular formula**
- ✓ The **same structural (bonding) formula**
- ✗ But a **different three-dimensional (spatial) arrangement** of atoms.

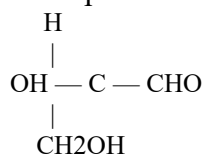
Stereoisomers are compounds that have the same structural formula but differ in the spatial arrangement of atoms.

💧 **Notice:** Monosaccharides exhibit stereoisomerism because they usually contain one or more **chiral (asymmetric) carbon atoms**.

Why Do Monosaccharides Show Stereoisomerism?

- ♦ Most monosaccharides have one or more **chiral (asymmetric)** carbon atoms.
- ♦ A **chiral carbon** is a carbon attached to **four different atoms or groups**.

Example:



Since all four substituents are different, this carbon is **chiral**.

🧠 Think of a chiral carbon like your **right and left hands**—they are mirror images but cannot be superimposed.

⚠ Exception: Dihydroxyacetone (DHA)

💧 Very High Yield

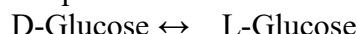
Dihydroxyacetone (DHA) is the **only monosaccharide that has no chiral carbon**.
Therefore:

- ✓ No D or L forms
- ✓ No enantiomers
- ✓ Not optically active
- 💡 **Clinical Pearl:** DHA is the **simplest ketose** and is **achiral**.

🌀 Enantiomers (Mirror-Image Isomers)

▽ **Enantiomers** are stereoisomers that are **non-superimposable mirror images** of each other.

Example:



They have:

- ✓ Same molecular formula
- ✓ Same chemical properties (in many non-biological settings)
- ✓ Mirror-image structures
- ✓ Different biological behavior

D and L Configuration

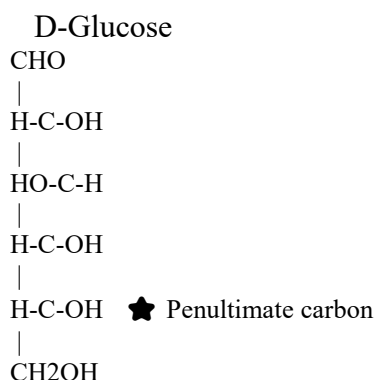
🔗 How Are D and L Sugars Determined?

🌀 Most Important Rule

Look at the **chiral carbon farthest from the carbonyl group** (the **penultimate carbon**) in the **Fischer projection**.

If the **–OH** is on the **RIGHT** ➡ **D-sugar**

If the **–OH** is on the **LEFT** ➡ **L-sugar**



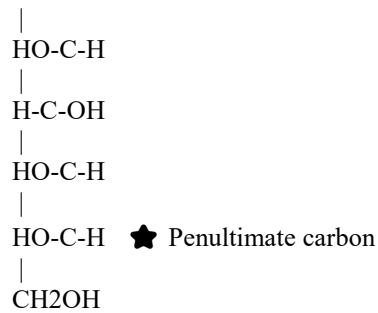
➡ **OH on the RIGHT**

☑ D-Glucose

[24/09/2026 10:40 AM] Imaanaa Harmee: L-Glucose

```

CHO
  
```



☞ **OH on the LEFT**

☒ L-Glucose

Reference Compound: Glyceraldehyde

Glyceraldehyde is the **reference carbohydrate** used to assign D and L configurations.

It has:

🕒 3 carbons

🕒 1 chiral carbon

🕒 2 enantiomers

Configuration OH Position

D-Glyceraldehyde Right

L-Glyceraldehyde Left

All other monosaccharides are compared with glyceraldehyde.

[24/09/2026 10:40 AM] Imaanaa Harmee: D ≠ Dextrorotatory (+)

Many students think:

D = rotates light to the right (+)

This is NOT always true.

Similarly:

L = rotates light to the left (−)

Also not always true.

Correct Concept

D/L describes the **configuration** (structure) relative to glyceraldehyde.

(+)/(−) describes the **direction of rotation** of plane-polarized light.

💡 **Example:**

D-Fructose is actually **levorotatory (−)**.

☞ **Notice:** D and L **do not predict** the direction of optical rotation.

Optical Activity

▽ A compound is **optically active** if it **rotates the plane of polarized light**.
This occurs because the molecule is **chiral**.

There are two possibilities:

➡ Dextrorotatory (+)

Rotates light **to the right (clockwise)**.

↩ Levorotatory (−)

Rotates light **to the left (counterclockwise)**.

Racemic Mixture

A **racemic (DL) mixture** contains **equal amounts of D and L enantiomers**.

Example:

50% D-glucose + 50% L-glucose

✂ The optical rotations cancel each other.

➡ **No net optical activity**

A racemic mixture is **optically inactive**, even though each enantiomer alone is optically active.

Number of Stereoisomers

Formula:-

♦ If a molecule has **n chiral carbons**: 2^n

Biological Importance

▽ Almost all naturally occurring sugars are **D-sugars**.

Examples:

✓ D-Glucose

✓ D-Fructose

✓ D-Galactose

Similarly:

▽ Most naturally occurring amino acids are **L-amino acids**.

[24/09/2026 10:40 AM] Imaanaa Harmee:

In biological systems, monosaccharides with five or more carbons exist predominantly (over 99%) in a **cyclic (ring) form** rather than an open-chain structure. This process occurs through an intramolecular reaction where the carbonyl group (the aldehyde or keto group) reacts with a hydroxyl group on the same sugar molecule.

The Anomeric Carbon

Cyclization creates a new asymmetric center at the former carbonyl carbon, which is designated as the **anomeric carbon**.

Aldoses: The anomeric carbon is **carbon 1**.

Ketoses: The anomeric carbon is **carbon 2**.

Ring Types

Depending on the size of the ring formed, the resulting cyclic structures are classified into two types:

Pyranose: A six-membered ring consisting of five carbons and one oxygen atom (e.g., glucopyranose).

Furanose: A five-membered ring consisting of four carbons and one oxygen atom (e.g., fructofuranose).

α and β Anomers

The creation of the anomeric carbon generates two different configurations, known as **anomers**, which are diastereomers.

α -configuration: In a Haworth projection, the hydroxyl (-OH) group on the anomeric carbon is drawn **down** (below the ring). In a modified Fischer projection, it is on the same side as the ring.

β -configuration: The hydroxyl group on the anomeric carbon is drawn **up** (above the ring).

Mutarotation

When cyclic anomers are in solution, they exist in equilibrium and can spontaneously (nonenzymatically) interconvert through the open-chain form in a process called **mutarotation**. While this happens spontaneously, cellular enzymes called **mutarotases** can accelerate the process.

However, if the hydroxyl group on the anomeric carbon forms a **glycosidic bond** with another molecule (such as in a disaccharide like lactose or a polysaccharide like glycogen), the configuration becomes **fixed** in either the α or β position, and mutarotation can no longer occur.

Reducing Sugars

A cyclized sugar is classified as a **reducing sugar** if the hydroxyl group on its anomeric carbon is not linked to another compound by a glycosidic bond. In this state, the ring can spontaneously open to its acyclic form, allowing the aldehyde group to be oxidized by chromogenic agents like **Benedict's reagent** or **Fehling's solution**. Glucose is a well-known

reducing sugar because its anomeric carbon remains free to undergo this oxidation.

[24/09/2026 10:40 AM] Imaanaa Harmee:

Derivatives of sugar

✓ Sugar derivatives are modified monosaccharides where functional groups are added, removed, or changed. These derivatives play critical roles in metabolism, structural integrity, and cellular signaling.

1. Reduced Sugars

Sugar reduction involves the gain of hydrogen atoms or the loss of an oxygen atom.

▽ **Polyols (Sugar Alcohols):** These are formed when the aldehyde or keto group of a sugar is reduced to a hydroxyl group, resulting in a molecule where all carbons contain alcohol groups.

▽ **Sorbitol (Glucitol):** Produced by the reduction of glucose via **aldose reductase**. In uncontrolled diabetes, sorbitol can accumulate in tissues like the lens and nerves, causing osmotic swelling and contributing to cataracts and neuropathy.

▽ **Galactitol:** Formed from the reduction of galactose. Its accumulation in the lens is a major factor in the formation of cataracts in patients with galactosemia.

▽ **Deoxysugars:** These occur when a hydroxyl group is replaced by a hydrogen atom. A primary biological example is **deoxyribose**, the five-carbon sugar found in the backbone of DNA.

2. Oxidized Sugars

Sugars can be oxidized at either the aldehyde carbon or the terminal hydroxyl carbon to form acids.

▽ **Aldonic Acids:** Formed by the oxidation of the aldehyde group at carbon 1. For example, glucose is oxidized to **gluconic acid** (or gluconate).

▽ **Uronic Acids:** Produced when the carbon containing the terminal hydroxyl group (carbon 6 in hexoses) is oxidized.

▽ **D-Glucuronic Acid:** A derivative of glucose used in **detoxification** reactions to make insoluble compounds like bilirubin, steroids, and certain drugs more water-soluble for excretion. It is also a key component of **glycosaminoglycans (GAGs)**.

▽ **L-Iduronic Acid:** The C-5 epimer of glucuronic acid, also essential for GAG structure.

3. Substituted Sugars

Monosaccharides often have their hydroxyl groups replaced or modified by other chemical

groups.

▽ **Sugar Phosphates:** Most intracellular sugars are phosphorylated at their terminal carbons (e.g., **glucose 6-phosphate**). This modification effectively traps the sugar inside the cell because there are no specific transporters for organic phosphates.

▽ **Amino Sugars:** A hydroxyl group (usually at carbon 2) is replaced by an amino group, such as in **glucosamine** or **galactosamine**.

▽ **N-Acetylated Amino Sugars:** The amino group is frequently acetylated to form derivatives like **N-acetylglucosamine (GlcNAc)** and **N-acetylgalactosamine (GalNAc)**. These are vital components of glycoproteins, glycolipids, and GAGs.

▽ **Sialic Acids:** A family of nine-carbon acidic sugars, the most common being **N-acetylneuraminic acid (NANA)**. They are often found at the terminal ends of oligosaccharide chains in glycoproteins and glycolipids.

4. Glycosides

Glycosides are formed when the hydroxyl group on the **anomeric carbon** of a sugar reacts with another compound.

▽ **O-Glycosidic Bonds:** Join sugars to other sugars (as in disaccharides like **lactose**) or to the hydroxyl groups of amino acids (serine or threonine) in **glycoproteins**.

▽ **N-Glycosidic Bonds:** Join a sugar to a nitrogenous base, as seen in **nucleosides** and **nucleotides** (e.g., the bond between ribose and adenine in **ATP**) or in N-linked glycoproteins where the sugar attaches to the amide nitrogen of asparagine.

[24/09/2026 10:40 AM] Imaanaa Harmee:

Disaccharides

✓ are carbohydrates composed of two monosaccharide units linked together by a covalent **glycosidic bond**. These molecules provide a significant portion of dietary calories and are primarily digested in the small intestine into their constituent simple sugars for absorption into the bloodstream.

I. Major Dietary Disaccharides

✓ Five primary disaccharides relevant to human nutrition and physiology:

♦ **Sucrose (Table Sugar):** Composed of **glucose and fructose** linked through an $\alpha(1\rightarrow2)$ **glycosidic bond**. It is the major natural sweetener found in fruits, honey, and molasses.

♦ **Lactose (Milk Sugar):** Consists of **galactose and glucose** joined by a $\beta(1\rightarrow4)$ **glycosidic bond**. It is the principal carbohydrate found exclusively in milk and dairy products.

♦ **Maltose:** A product of the enzymatic digestion of polysaccharides (like starch), composed of **two glucose units** linked via an $\alpha(1\rightarrow4)$ **glycosidic bond**. It is found in significant quantities in beer and malt liquors.

♦ **Isomaltose:** Also a product of starch digestion, it consists of **two glucose units** linked by an $\alpha(1\rightarrow6)$ **glycosidic bond**, which corresponds to the branch points of amylopectin and glycogen.

♦ **Trehalose:** A disaccharide of **two glucose units** joined by an unusual $\alpha(1\rightarrow1)$ **bond** between their anomeric carbons. It is found in mushrooms, fungi, algae, and insects.

II. Chemical Properties and Linkages

✓ Disaccharides are characterized by the specific nature of their **O-glycosidic bonds**, which form between the hydroxyl group of the **anomeric carbon** of one sugar and a hydroxyl group of another.

▽ **Naming Linkages:** Bonds are named according to the carbon numbers involved and the configuration (α or β) of the anomeric hydroxyl group of the first sugar in the bond.

▽ **Reducing vs. Non-reducing Sugars:** A disaccharide is a **reducing sugar** if the anomeric carbon of its second monosaccharide is not involved in a glycosidic bond, allowing that ring to open and the aldehyde group to be oxidized.

▽ **Lactose and maltose** are reducing sugars. In contrast, **sucrose and trehalose** are **non-reducing** because their glycosidic bonds link the anomeric carbons of both monosaccharides, preventing the rings from opening.

III. Digestion and Absorption

✓ Disaccharides cannot be absorbed directly; they must be hydrolyzed into monosaccharides by specific enzymes known as **disaccharidases**.

▽ **Localization:** These enzymes are glycoproteins located on the **brush border membrane** (microvilli) of the intestinal mucosal cells, primarily in the duodenum and upper jejunum.

▽ **Specific Enzymes:**

☞ **Lactase (β -galactosidase):** Hydrolyzes lactose into glucose and galactose.

☞ **Sucrase:** Cleaves sucrose into glucose and fructose.

☞ **Maltase:** Cleaves maltose into two glucose units.

☞ **Isomaltase:** Hydrolyzes the $\alpha(1\rightarrow6)$ bond of isomaltose.

☞ **Trehalase:** Cleaves the $\alpha(1\rightarrow1)$ bond of trehalose.

Enzyme Complexes: Sucrase and isomaltase are subunits of a single bifunctional protein known as the **sucrase-isomaltase complex**.

IV. Synthesis: Lactose as a Special Case

Unlike other dietary disaccharides, lactose is synthesized by the human body in the **mammary glands** of lactating females. This is catalyzed by **lactose synthase**, an enzyme complex composed of two proteins:

△ **Protein A (β -D-galactosyltransferase):** Found in many tissues, it normally synthesizes N-acetyllactosamine for glycoproteins.

△ **Protein B (α -lactalbumin):** Found only in lactating mammary glands; its synthesis is stimulated by the hormone **prolactin**. When Protein B binds to Protein A, it changes the enzyme's specificity so that it uses glucose as a substrate to produce lactose.

[24/09/2026 10:40 AM] Imaanaa Harmee:

Reducing vs. non-reducing sugars

▽ A sugar is classified as **reducing** or **non-reducing** based on the status of its **anomeric carbon**—the carbon that was part of the carbonyl group (aldehyde or keto) before the sugar cyclized.

Reducing Sugars

✓ A sugar is defined as a reducing sugar if the hydroxyl group on its anomeric carbon is **not linked** to another compound by a glycosidic bond.

🧬 **Mechanism:** Because the anomeric hydroxyl is free, the cyclic ring can spontaneously open into its acyclic (straight-chain) form. In this state, the free aldehyde group can act as a reducing agent, donating electrons to other compounds.

🧬 **Testing:** These sugars react with chromogenic agents such as **Benedict's reagent**, **Fehling's solution**, or **Clinitest**. During the reaction, the sugar's aldehyde group is oxidized into a carboxyl group while the copper in the reagent is reduced, producing a characteristic color change.

⚙️ **A Note on Ketoses:** Although fructose is a ketose (containing a keto group rather than an aldehyde), it still reacts as a reducing sugar in alkaline test solutions because it forms **tautomers** that are aldehydes.

Examples:

✓ **All monosaccharides:** Glucose, fructose, galactose, and xylulose.

✓ **Specific disaccharides:** **Lactose** and **maltose**. Lactose is reducing because the anomeric end of its glucose residue is not involved in the glycosidic linkage.

Non-reducing Sugars

▽ A sugar is non-reducing if its anomeric carbon is **covalently attached** to another molecule through a glycosidic bond.

🧬 **Mechanism:** Once the anomeric carbon forms a glycosidic bond, the ring is "locked" and can no longer open to form a straight-chain aldehyde or ketone. Without a free carbonyl group, the sugar cannot donate electrons to reduce reagents like copper.

🧬 **Testing:** These sugars do not yield a positive result with the Clinitest or similar copper-reduction methods.

Examples:

✓ **Sucrose:** This disaccharide is non-reducing because the glycosidic bond joins the anomeric carbons of both glucose and fructose, leaving no free anomeric hydroxyl groups.

✓ **Trehalose:** A disaccharide where two glucose units are linked through their anomeric

carbons.

✓ **Sugar Alcohols (Polyols):** Compounds like **sorbitol** and **galactitol** are not reducing sugars because their aldehyde group has been reduced to a hydroxyl group, meaning they no longer possess a carbonyl group to oxidize.

[24/09/2026 10:40 AM] Imaanaa Harmee:

OLIGOSACCHARIDE SUGARS

✓ **Oligosaccharides** are carbohydrates composed of roughly **three to ten** (or sometimes up to twelve) monosaccharide units joined together by covalent **glycosidic bonds**. They occupy a middle ground in complexity between simple disaccharides (two units) and long-chain polysaccharides (more than ten units).

I. Role in Digestion

Oligosaccharides often appear as intermediate products during the breakdown of larger polysaccharides like starch and glycogen.

☒ **Dextrins:** When salivary or pancreatic α -**amylase** acts on starch, it hydrolyzes random $\alpha(1\rightarrow4)$ bonds, producing a mixture of short, branched, and unbranched oligosaccharides known as dextrins.

☒ **Limit Dextrins:** Because α -amylase cannot hydrolyze the $\alpha(1\rightarrow6)$ bonds at branch points, the resulting fragments—typically **four to nine glucosyl residues** long—are called limit dextrins.

☒ **Maltotriose:** A specific trisaccharide consisting of three glucose units produced during the digestion of starch by α -amylase.

II. Structural and Biological Functions

In the human body, oligosaccharides are rarely free-floating; they are most commonly found as components of **complex carbohydrates** attached to other non-carbohydrate structures.

☒ **Glycoproteins:** Most globular proteins in human plasma and many membrane proteins are glycoproteins, which contain relatively short (usually **two to ten residues**), often branched oligosaccharide chains.

☒ **N- and O-Linkages:** These chains can be attached to the amide nitrogen of **asparagine** (N-linked) or the hydroxyl group of **serine, threonine, or hydroxylysine** (O-linked).

☒ **Cellular Signaling:** Surface oligosaccharides are vital for **cell-surface recognition** (by hormones or viruses), **antigenicity** (such as the ABO blood group determinants), and protecting cells from proteolysis.

☒ **Dolichol-linked Oligosaccharide:** During the synthesis of N-linked glycoproteins, a specialized **14-sugar oligosaccharide** (containing N-acetylglucosamine, mannose, and glucose) is first assembled on a lipid called dolichol before being transferred to the protein.

III. Dietary Sources and Malabsorption

☒ **Legumes:** Beans and peas contain oligosaccharides with linked **galactose residues** (such as sucrose with one to three additional galactoses) that humans lack the enzymes to hydrolyze.

☒ **Bacterial Fermentation:** Because these specific oligosaccharides cannot be absorbed in the small intestine, they pass into the **colon**. There, bacteria metabolize them into gases (H_2 , CO_2 , CH_4) and short-chain fatty acids, often resulting in flatulence and abdominal distension.

[24/09/2026 10:40 AM] Imaanaa Harmee:

Polysaccharides

▽ are complex carbohydrates composed of more than ten monosaccharide units, and they can reach lengths of hundreds or even thousands of sugar units.

▽ Unlike simple sugars, they often do not have a sweet taste. They are classified based on their structure (linear or branched) and their composition (homopolysaccharides, containing one type of sugar, or heteropolysaccharides, containing multiple types).

I. Major Storage Polysaccharides

Storage polysaccharides serve as energy reservoirs that can be mobilized when the body requires glucose.

Glycogen: This is the primary **storage form of glucose in animals**, found mainly in the liver and skeletal muscle.

Structure: It is a highly branched homopolymer made exclusively of α -D-glucose.

Linkages: The primary glycosidic bond is an $\alpha(1 \rightarrow 4)$ **linkage**, with branches occurring every eight to ten residues via $\alpha(1 \rightarrow 6)$ **linkages**.

Function: Liver glycogen maintains blood glucose levels during fasting, while muscle glycogen provides fuel for ATP synthesis during contraction.

Starch: The **storage form of glucose in plants**, providing a major source of dietary calories for humans. It exists in two forms:

Amylose: A linear, unbranched chain of glucose residues linked by $\alpha(1 \rightarrow 4)$ **bonds**.

Amylopectin: A branched polymer similar to glycogen but typically less highly branched, containing both $\alpha(1 \rightarrow 4)$ **and** $\alpha(1 \rightarrow 6)$ **linkages**.

II. Structural and Functional Polysaccharides

These polysaccharides provide physical integrity to cells and tissues or perform specialized physiological roles.

▽ **Cellulose:** A structural component of plant cell walls.

▽ **Structure:** It is an unbranched polymer of glucose residues joined by $\beta(1 \rightarrow 4)$ **glycosidic bonds**.

▽ **Indigestibility:** Humans do not produce $\beta(1 \rightarrow 4)$ -endoglucosidase, meaning we **cannot digest cellulose**. It passes through the digestive tract as dietary fiber.

▽ **Glycosaminoglycans (GAGs):** These are long, unbranched **heteropolysaccharides** composed of repeating disaccharide units, usually consisting of an acidic sugar and an amino sugar.

▽ **Properties:** GAGs are **highly negatively charged**, allowing them to bind large amounts of water and form a gel-like matrix.

Examples: Common GAGs include **hyaluronic acid** (a lubricant in joints), **chondroitin sulfate** (found in cartilage and tendons), and **heparin** (an intracellular anticoagulant).

Proteoglycans: Most GAGs (except hyaluronic acid) are covalently attached to a core protein, forming structures that resemble a "bottle brush".

IV. Dietary Fiber

Dietary fiber consists principally of plant polysaccharides (like cellulose and hemicellulose) and lignin that remain **undigested by human enzymes**. While they provide little energy, they add bulk to the diet, delay gastric emptying, and can be fermented by colonic bacteria into short-chain fatty acids and gases (CO_2 , H_2 , and methane).

[24/09/2026 10:40 AM] Imaanaa Harmee:

Proteoglycans

✓ are large complexes consisting of a **core protein** to which long, unbranched heteropolysaccharide chains called **glycosaminoglycans (GAGs)** are covalently attached. Unlike glycoproteins, which are primarily protein, proteoglycans typically consist of **over 95% carbohydrate**.

Structure and Composition

▽ **The "Bottle Brush" Model:** Because GAG chains are highly negatively charged, they repel one another and radiate out from the core protein. This creates a structure resembling a **bottle brush**, where the GAGs are the bristles and the protein is the wire core.

▽ **Glycosaminoglycans (GAGs):** These are the carbohydrate components of proteoglycans, composed of **repeating disaccharide units** [acidic sugar–amino sugar]_n.

▽ **Amino Sugars:** Usually D-glucosamine or D-galactosamine, which are often **acetylated** to eliminate positive charge and frequently **sulfated**.

▽ **Acidic Sugars:** Generally D-glucuronic acid or its C-5 epimer, L-iduronic acid.

▽ **Linkage Region:** The GAG chains are most commonly linked to the core protein through a **trihexoside (galactose-galactose-xylose)** and a **serine residue**. An **O-glycosidic bond** forms between the xylose and the hydroxyl group of the serine.

▽ **Proteoglycan Aggregates:** Many proteoglycan monomers associate noncovalently with a single long molecule of **hyaluronic acid** to form massive aggregates, which are stabilized by small **link proteins**.

Physical Properties and Function

◆ **Hydration and Lubrication:** Due to their high negative charge, proteoglycans bind large

amounts of water, producing a **gel-like matrix**. This makes them essential as biological lubricants and shock absorbers.

♦**Resilience:** When proteoglycans are compressed, water is "squeezed out" into a smaller volume. When the pressure is released, the negative charges repel each other, causing the molecule to spring back to its original hydrated volume.

♦**Location:** They are vital components of the **extracellular matrix (ECM)**, cartilage, tendons, ligaments, the vitreous humor of the eye, and mucous secretions.

Synthesis and Degradation

◇**Synthesis:** Proteoglycans are synthesized primarily in the **Golgi apparatus**. The process involves the sequential addition of alternating acidic and amino sugars donated by **UDP-derivatives** (such as UDP-glucuronic acid and UDP-N-acetylglucosamine). The final step is often **sulfation**, which requires the sulfate donor **PAPS**.

◇**Degradation:** They are degraded within **lysosomes** by a variety of **acid hydrolases**. This occurs sequentially from the nonreducing end of the chain—a "last on, first off" process.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Glycosaminoglycans (GAGs)

✓ formerly known as mucopolysaccharides, are large complexes composed of long, unbranched heteropolysaccharide chains. These molecules are a vital component of the body's ground substance and the **extracellular matrix (ECM)**, where they interact with fibrous proteins like collagen and elastin.

1. Structure and Composition

✓ GAGs are generally composed of a **repeating disaccharide unit** [acidic sugar–amino sugar]_n:

△ **Amino Sugar:** Usually **D-glucosamine** or **D-galactosamine**. The amino group is typically acetylated, which eliminates its positive charge, and is frequently sulfated on carbon 4 or 6.

△ **Acidic Sugar:** Generally **D-glucuronic acid** or its C-5 epimer, **L-iduronic acid**.

☞**Exception:** **Keratan sulfate** is the only major GAG that contains galactose instead of an acidic sugar.

△ **Negative Charge:** The presence of carboxyl groups on acidic sugars and sulfate groups on amino sugars gives GAGs a **strongly negative nature**.

2. Physical Properties and Function

☉**Hydration and Support:** Due to their high negative charge, GAG chains repel one another and attract a shell of water molecules. This allow them to bind large amounts of

water, producing a **gel-like matrix** that serves as flexible support for the ECM.

☉ **Resilience and Lubrication:** In solution, GAGs "slip" past one another, producing the "slippery" consistency of **synovial fluid** and mucous secretions. When compressed, water is "squeezed out," but when the pressure is released, the negative charges repel each other, causing the GAGs to spring back to their original hydrated volume.

☉ **Proteoglycans:** With the exception of hyaluronic acid, all GAGs are found covalently attached to a core protein, forming **proteoglycan monomers**. These monomers often resemble a "**bottle brush**," with GAG chains radiating out from the protein core.

3. Major Classes of GAGs

The six major classes are categorized by their monomeric composition and distribution:

♦ **Chondroitin 4- and 6-sulfates:** The most abundant GAGs in the body; found in cartilage, tendons, and ligaments.

♦ **Hyaluronic Acid (Hyaluronate):** Unique because it is **unsulfated**, not covalently attached to a protein, and found in both animal tissues and bacteria. It serves as a lubricant and shock absorber in joints and the vitreous humor.

♦ **Keratan Sulfates (I and II):** Found in the cornea (KS I) and loose connective tissue (KS II).

♦ **Dermatan Sulfate:** Found in the skin, blood vessels, and heart valves.

♦ **Heparan Sulfate:** An extracellular GAG found in basement membranes and on cell surfaces.

♦ **Heparin:** Unlike other GAGs, it is primarily **intracellular**, found in mast cells, and serves as an **anticoagulant**.

4. Synthesis and Degradation

⚗ **Synthesis:** Occurs primarily in the **Golgi apparatus**. Chains are elongated by glycosyltransferases using **UDP-derivatives** as sugar donors. The final step is **sulfation**, using **PAPS** (3'-phosphoadenosyl-5'-phosphosulfate) as the sulfate donor.

⚗ **Degradation:** GAGs are degraded in **lysosomes** by **acid hydrolases**. Digestion is a sequential process starting from the nonreducing end—a "**last on, first off**" mechanism where the last group added during synthesis is the first removed.

5. Clinical Relevance

Hereditary deficiencies in any one of the lysosomal hydrolases required for GAG degradation result in **mucopolysaccharidoses**. These progressive disorders lead to the accumulation of GAGs in various tissues, causing skeletal deformities and mental retardation. Examples include **Hunter** and **Hurler syndromes**. Additionally, defects in the sulfation of GAGs can result in **chondrodystrophies**, affecting skeletal maintenance.

[24/09/2026 10:41 AM] Imaanaa Harmee: ☉ The six major classes of GAGs are categorized by their monomeric composition, the type of glycosidic linkages, and the degree and location of sulfate groups.

1. Chondroitin 4- and 6-Sulfates

▽ **Composition:** These consist of N-acetylgalactosamine (sulfated on either C-4 or C-6) and glucuronic acid.

▽ **Distribution and Function:** They are the **most abundant GAGs in the body**. They are found primarily in **cartilage, tendons, ligaments, and the aorta**. In cartilage, they play a critical role by binding to collagen to hold fibers in a strong, tight network.

2. Hyaluronic Acid (Hyaluronate)

▽ **Composition:** Composed of N-acetylglucosamine and glucuronic acid.

▽ **Unique Features:** Unlike all other major GAGs, hyaluronic acid is **unsulfated** and is **not covalently attached to a protein core**. It is also the only GAG found in both animal tissues and bacteria.

▽ **Distribution and Function:** It serves as a vital **lubricant and shock absorber**. It is found in the synovial fluid of joints, the vitreous humor of the eye, the umbilical cord, and loose connective tissue.

3. Keratan Sulfates (KS I and KS II)

▽ **Composition:** These are unique because they **lack an acidic sugar**, containing galactose instead, paired with N-acetylglucosamine. They are the most heterogeneous GAGs as they contain additional monosaccharides like L-fucose and mannose.

▽ **Distribution:** **KS I** is found primarily in the **cornea**, while **KS II** is found in **loose connective tissue** proteoglycan aggregates along with chondroitin sulfate.

4. Dermatan Sulfate

▽ **Composition:** Composed of N-acetylgalactosamine and **L-iduronic acid** (with variable amounts of glucuronic acid).

▽ **Distribution:** This class is located in the **skin, blood vessels, and heart valves**.

5. Heparan Sulfate

▽ **Composition:** It contains the same disaccharide unit as heparin (glucosamine and glucuronic/iduronic acid), but features more N-acetylated glucosamines and fewer sulfate groups.

▽ **Distribution:** It is an **extracellular GAG** found in basement membranes and is a ubiquitous component of cell surfaces.

6. Heparin

▽ **Composition:** Primarily composed of glucosamine (most bound in sulfamide linkages)

and glucuronic or iduronic acid. It is highly sulfated, with an average of 2.5 sulfate groups per disaccharide unit.

▽ **Unique Features:** Unlike other GAGs, which are extracellular, heparin is an **intracellular component**.

▽ **Distribution and Function:** It is found within **mast cells** that line arteries, particularly in the liver, lungs, and skin. Its primary physiological role is to serve as an **anticoagulant**.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Proteoglycans

▽ are large molecular complexes consisting of a **core protein** covalently attached to long, unbranched heteropolysaccharide chains known as **glycosaminoglycans (GAGs)**. They are distinguished from glycoproteins by their carbohydrate content, which typically accounts for **over 95% of their mass**.

Structure and Composition

▽ **The "Bottle Brush" Model:** Because the GAG chains are highly negatively charged, they repel one another and radiate out from the core protein. This creates a structure resembling a bottle brush, where the protein core is the central "wire" and the GAGs are the radiating "bristles".

▽ **Glycosaminoglycans (GAGs):** These are the carbohydrate components, generally made of **repeating disaccharide units** [acidic sugar–amino sugar]_n.

▽ **Acidic Sugars:** Typically D-glucuronic acid or its C-5 epimer, L-iduronic acid (except in keratan sulfate, which contains galactose instead).

▽ **Amino Sugars:** Either D-glucosamine or D-galactosamine, which are usually acetylated (to eliminate positive charge) and frequently sulfated.

▽ **Linkage Region:** GAG chains are most commonly joined to the core protein through a **trihexoside (galactose-galactose-xylose)** attached to a **serine residue** via an **O-glycosidic bond**.

▽ **Proteoglycan Aggregates:** Proteoglycan monomers can associate noncovalently with a single long molecule of **hyaluronic acid** to form massive aggregates. These structures are stabilized by small **link proteins**.

Physical Properties and Function

◆**Hydration and Support:** Due to their high negative charge, proteoglycans repel each other and attract a shell of water molecules. This creates a **gel-like matrix** that forms the basis of the body's ground substance and provides flexible support for the extracellular matrix (ECM).

♦**Resilience and Lubrication:** In solution, proteoglycans "slip" past each other, giving mucous secretions and synovial fluid a "slippery" consistency. When compressed, water is squeezed out, but the repulsion of negative charges causes them to spring back to their original hydrated volume once pressure is released, allowing them to act as **shock absorbers**.

♦**Distribution:** They are vital components of the ECM, cartilage, tendons, ligaments, the aorta, the vitreous and aqueous humor of the eye, and mucous secretions.

Synthesis and Degradation

♦**Synthesis:** The core protein is synthesized in the rough endoplasmic reticulum (RER) and subsequently glycosylated in the **Golgi apparatus**. Specific glycosyltransferases sequentially add acidic and amino sugars donated by **UDP-derivatives** (such as UDP-xylose or UDP-glucuronic acid).

♦**Degradation:** Proteoglycans are degraded within **lysosomes** by a series of **acid hydrolases**. The process is sequential, moving from the nonreducing end of the chain in a "last on, first off" manner.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Peptidoglycan

✂ is a unique protein–polysaccharide structure that forms the **cell wall** of bacteria, located exterior to their plasma membrane. It serves as a vital **protective shield** for the bacterial cell.

1. Structure and Composition

✓ **Polysaccharide Chains:** The core structure consists of extensively **cross-linked polysaccharides**.

✓ **Cross-linking:** These polysaccharide chains are held together by peptide cross-links. A unique feature of these cross-links is the presence of **D-amino acids**, which are not found in human proteins but serve critical functions in the bacterial cell wall.

🔗 Distinction from Proteoglycans

While the names are similar, peptidoglycans and **proteoglycans** are distinct:

▽ **Peptidoglycans** are specific to **bacterial cell walls** and consist of cross-linked sugars and peptides.

▽ **Proteoglycans** are found in the **human extracellular matrix** (e.g., cartilage) and consist of a core protein with long, unbranched glycosaminoglycan (GAG) chains.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Glycoproteins

▽ are proteins containing covalently attached oligosaccharide chains.

▽ They are distinguished from proteoglycans by having relatively **short carbohydrate chains** (typically 2 to 10 sugar residues) that are often **branched** and do not contain serial repeating disaccharide units.

▽ While some glycoproteins like immunoglobulin IgG contain less than 4% carbohydrate, others like mucin can be over 80% carbohydrate by mass.

I. Structure and Linkages

The oligosaccharides in glycoproteins are attached to the protein through two primary types of covalent bonds:

✓ **N-glycosidic link:** The carbohydrate is attached to the **amide group of an asparagine** side chain. N-linked oligosaccharides are further classified into "complex" (containing diverse sugars like fucose and NANA) or "high-mannose".

✓ **O-glycosidic link:** The carbohydrate is attached to the **hydroxyl group of serine or threonine** (or hydroxylysine in the case of collagen).

II. Biological Functions

Almost all globular proteins in human plasma are glycoproteins. Their functions include:

✓ **Cell Surface Phenomena:** They mediate cell-surface recognition (by hormones, other cells, or viruses) and provide **cell surface antigenicity**, such as the ABO blood group determinants.

✓ **Protection and Lubrication: Mucins** are glycoproteins that serve as protective biological lubricants in the gastrointestinal and urogenital tracts.

✓ **Extracellular Matrix (ECM):** They are essential components of the ECM and help protect cell surfaces from proteolysis or immune attack.

III. Synthesis and Targeting

Synthesis occurs on ribosomes attached to the **rough endoplasmic reticulum (RER)**, where an N-terminal signal sequence directs the growing polypeptide into the RER lumen.

✓ **N-linked synthesis:** A 14-sugar oligosaccharide is first assembled on **dolichol** (a specialized ER membrane lipid) before being transferred as a whole unit to an asparagine residue.

✓ **O-linked synthesis:** Sugars are added sequentially, one at a time, from nucleotide carriers primarily within the **Golgi apparatus**.

✓ **Targeting:** The Golgi complex acts as a sorting center. For example, enzymes destined

for **lysosomes** are specifically phosphorylated at mannose residues to create a **mannose 6-phosphate** signal.

IV. Clinical Relevance

I-Cell Disease: A rare lysosomal storage disease caused by a deficiency in the ability to phosphorylate mannose residues. Without the mannose 6-P signal, lysosomal acid hydrolases are incorrectly targeted to extracellular sites rather than the lysosomes.

Glycoprotein Storage Diseases (Oligosaccharidoses): Caused by a deficiency in any one of the lysosomal acid hydrolases required to degrade glycoproteins sequentially ("last on, first off").

Nonenzymatic Glycation: In conditions like diabetes, high blood glucose levels lead to the **nonenzymatic** attachment of glucose to proteins (such as hemoglobin to form **HbA1c**). This process can distort protein structure and is a major contributor to diabetic complications like retinopathy and nephropathy.

[24/09/2026 10:41 AM] Imaanaa Harmee:
Lipid

[24/09/2026 10:41 AM] Imaanaa Harmee:
Definition of Lipids

▽ **Lipids are a heterogeneous group of structurally diverse organic molecules that are defined by their hydrophobicity (water-insolubility).** They are generally insoluble in aqueous solutions but can be extracted from tissues using **nonpolar solvents**.

General Properties of Lipids

▽ **Hydrophobicity:** The defining characteristic of lipids is that they are relatively insoluble in water.

▽ **Amphipathic Nature:** While many lipids are purely hydrophobic, some, such as **phospholipids** and **fatty acids**, are amphipathic, meaning they possess both a hydrophilic (polar) region and a hydrophobic (nonpolar) region.

▽ **Compartmentalization:** Due to their insolubility in water, lipids in the body are typically compartmentalized, such as being packed into **membranes**, stored as droplets of **triacylglycerol** in adipocytes, or transported in the blood associated with proteins in **lipoprotein particles** or on **albumin**.

▽ **Energy Density:** Lipids are highly reduced and largely anhydrous, providing a concentrated store of energy. The complete oxidation of triacylglycerols releases approximately **9 kcal/g**, which is more than twice the energy yield of carbohydrates or proteins.

Biological Functions of Lipids

Lipids serve a wide variety of essential roles in the human body:

▽ **Major Energy Source and Reserve:** Lipids are a primary fuel for many tissues,

particularly the liver and muscle. Triacylglycerols stored in white adipose tissue serve as the body's **major energy reserve**.

▽ **Structural Components of Membranes:** Lipids, including phospholipids, glycolipids, and cholesterol, provide the **hydrophobic barrier** that permits the partitioning of the aqueous contents of cells and subcellular organelles.

▽ **Membrane Fluidity Regulation: Cholesterol** is a vital structural component of all cell membranes, where it functions to modulate and maintain membrane fluidity.

▽ **Precursors for Bioactive Molecules:** **Cholesterol** is the precursor for **bile acids**, **steroid hormones** (such as cortisol, estrogen, and testosterone), and **Vitamin D**.

Essential fatty acids (linoleic and -linolenic acid) are precursors for **prostaglandins**, **thromboxanes**, and **leukotrienes**, which act as local mediators of inflammation and other physiological processes.

▽ **Vitamins and Coenzymes:** Fat-soluble vitamins (**A, D, E, and K**) have critical regulatory or coenzyme functions. For example, Vitamin K is essential for blood clotting.

▽ **Intracellular Signaling:** Certain membrane phospholipids, such as **phosphatidylinositol (PI)**, serve as reservoirs for intracellular messengers like IP and diacylglycerol (DAG) that mediate cellular responses to hormones and neurotransmitters.

▽ **Specialized Protective Roles:**

✓ **Lung surfactant** (primarily dipalmitoylphosphatidylcholine) reduces surface tension in the alveoli to prevent lung collapse.

✓ **Sphingomyelin** is an essential component of the **myelin sheath** that insulates and protects nerve fibers.

✓ **Bile salts** (derivatives of cholesterol) act as detergents to emulsify dietary lipids, facilitating their digestion and absorption.

[24/09/2026 10:41 AM] Imaanaa Harmee:

simple lipids

◆ are primarily defined as esters of fatty acids with various alcohols. The most significant simple lipids discussed in the context of human physiology are **fatty acids** and **triacylglycerols** (also known as neutral fats).

1. Fatty Acids

Fatty acids are the basic building blocks of most lipids. They consist of a hydrophobic hydrocarbon chain with a terminal hydrophilic carboxyl group.

✓ **Saturation Levels:**

▽ **Saturated Fatty Acids:** Contain no double bonds in the carbon chain (e.g., palmitic and stearic acids).

▽ **Unsaturated Fatty Acids:** Contain one or more double bonds. They are classified as **monounsaturated** (one double bond, like oleic acid) or **polyunsaturated** (two or more double bonds, like linoleic acid).

✓ **Isomerism (Cis vs. Trans):** In nature, the double bonds of unsaturated fatty acids are nearly always in the **cis configuration**, which creates a "kink" in the chain and helps maintain membrane fluidity. **Trans fatty acids** are primarily produced industrially through the hydrogenation of vegetable oils and are associated with increased risk of heart disease.

✓ **Essential Fatty Acids:** Humans cannot synthesize fatty acids with double bonds near the methyl () end. Therefore, **linoleic acid** (-6) and **-linolenic acid** (-3) must be obtained from the diet.

2. Acylglycerols (Neutral Fats)

Acylglycerols are formed by the esterification of fatty acids to a **glycerol** molecule.

✓ **Triacylglycerols (Triglycerides):** These are the most common simple lipids and consist of **three fatty acids** esterified to one glycerol moiety.

✓ **Chemical Nature:** Because the negative charge of the carboxyl group is lost during esterification, these molecules are often referred to as "**neutral fats**".

✓ **Energy Storage:** Triacylglycerols are the body's **major energy reserve**. They are stored as concentrated, anhydrous (water-free) droplets within **adipocytes** (fat cells).

✓ **Caloric Yield:** Upon complete oxidation, triacylglycerols release approximately **9 kcal/g**, which is more than double the energy provided by carbohydrates or proteins (4 kcal/g).

3. Physical States (Fats vs. Oils)

The physical state of simple lipids at room temperature depends on their fatty acid composition:

✓ **Fats:** Solid at room temperature; usually contain a high proportion of **saturated fatty acids** (common in animal products).

✓ **Oils:** Liquid at room temperature; usually contain a high proportion of **unsaturated fatty acids** (common in plants and fish).

[24/09/2026 10:41 AM] Imaanaa Harmee:

compound (complex) lipids

✓ are heterogeneous groups of water-insoluble organic molecules that contain other chemical groups in addition to fatty acids and an alcohol. These molecules are predominantly found in cell membranes, where they provide structural integrity and facilitate intercellular

communication.

✓ The two primary categories of complex lipids are **phospholipids** and **glycolipids**, along with **lipoproteins**, which serve as the transport vehicles for these and other lipids in the bloodstream.

I. Phospholipids

Phospholipids are polar, ionic, amphipathic compounds composed of an alcohol attached by a **phosphodiester bridge** to either **diacylglycerol** or **sphingosine**. They are the predominant lipids of cell membranes.

1. Glycerophospholipids (Phosphoglycerides)

These contain a **glycerol backbone** and are all derivatives of **phosphatidic acid** (diacylglycerol with a phosphate group on carbon 3).

Common Types: Depending on the alcohol attached to the phosphate group, they include:

Phosphatidylcholine (Lecithin): The major phospholipid in most cell membranes and a component of lung surfactant.

Phosphatidylethanolamine and Phosphatidylserine: Key structural membrane components.

Phosphatidylinositol (PI): Serves as a reservoir for arachidonic acid (a prostaglandin precursor) and acts as an intracellular signaling molecule.

Specialized Types:

Cardiolipin: Found almost exclusively in the **inner mitochondrial membrane**, it is essential for the function of the electron transport chain.

Plasmalogens: Ether lipids where the fatty acid at carbon 1 is replaced by an unsaturated alkyl group; they are abundant in nerve and heart tissue.

2. Sphingophospholipids

These are built on a **sphingosine** (an amino alcohol) backbone rather than glycerol.

Sphingomyelin: The only significant sphingophospholipid in humans. It is formed by attaching a fatty acid to sphingosine (forming **ceramide**) and adding phosphorylcholine. It is a vital component of the **myelin sheath** that insulates nerve fibers.

II. Glycolipids (Glycosphingolipids)

Glycolipids contain both carbohydrate and lipid components. They are derivatives of **ceramides** where a sugar (monosaccharide or oligosaccharide) is attached directly to the ceramide by an O-glycosidic bond. They contain no phosphate.

1. Neutral Glycosphingolipids

Cerebrosides: The simplest neutral glycolipids, containing a single sugar molecule (glucose

or galactose). They are abundant in the brain and peripheral nervous tissue.

Globosides: Contain neutral oligosaccharides attached to ceramide.

2. Acidic Glycosphingolipids

These are negatively charged at physiologic pH.

Gangliosides: The most complex glycosphingolipids, found primarily in the ganglion cells of the CNS. They contain one or more molecules of **N-acetylneuraminic acid (NANA)**, which provides the negative charge.

Sulfatides: Cerebrosides containing a sulfated galactosyl residue; they are found primarily in nerve tissue and the kidneys.

III. Lipoproteins

Lipoproteins are spherical macromolecular complexes of lipids and specific proteins (**apolipoproteins**). They function to keep hydrophobic lipids soluble and transport them through the aqueous environment of the plasma to various tissues.

Composition: They consist of a neutral lipid core (triacylglycerols and cholesteryl esters) surrounded by a shell of amphipathic apolipoproteins, phospholipids, and unesterified cholesterol.

Classification: Classified by size and density into **chylomicrons**, **VLDL** (very-low-density), **LDL** (low-density), and **HDL** (high-density) lipoproteins.

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IV. Clinical Significance

Deficiencies in the enzymes required to degrade complex lipids lead to serious medical conditions:

Sphingolipidoses: Lysosomal storage diseases where sphingolipids accumulate due to missing hydrolases. Examples include **Tay-Sachs**, **Gaucher**, and **Niemann-Pick** diseases.

Respiratory Distress Syndrome (RDS): Caused by a deficiency in the glycerophospholipid **dipalmitoylphosphatidylcholine (DPPC)**, the main component of lung surfactant.

Atherosclerosis: Linked to the gradual deposition of cholesterol carried by lipoproteins (particularly **LDL**) in the arterial walls.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Fatty acids

✔ are long-chain aliphatic hydrocarbon structures with a terminal carboxyl group that play essential roles as the body's primary fuel source, structural components of cell membranes, and precursors for signaling molecules.

I. Structure and Physicochemical Properties

A fatty acid consists of a **hydrophobic hydrocarbon chain** and a **hydrophilic carboxyl**

group.

Ionization: The carboxyl group has a pKa of approximately 4.8. At physiologic pH (~7.4), this group is dissociated, forming the negatively charged **carboxylate ion** (-COO^-).

Amphipathic Nature: This dual nature (polar head and nonpolar tail) makes fatty acids **amphipathic**. However, in **long-chain fatty acids (LCFAs)**, the hydrophobic portion is so predominant that the molecules remain highly **water-insoluble**.

Solubility and Transport: Due to their insolubility, unesterified (free) fatty acids must be transported in the blood associated with **serum albumin**. Esterified fatty acids are transported within **lipoprotein particles** such as chylomicrons and VLDL.

II. Chain Length and Saturation

Fatty acids are classified by the length of their carbon chain and the presence of double bonds.

Chain Length: Most naturally occurring fatty acids in humans have an **even number of carbons**, typically between 16 and 20.

Short- and Medium-chain: Fatty acids with 4 to 10 carbons are found in significant quantities in **milk fat**.

Very-long-chain (VLCFA): The brain can produce fatty acids longer than 22 carbons for specialized lipids.

Saturation Levels:

Saturated Fatty Acids: Contain **no double bonds** between carbons. Common examples include **palmitic acid (16:0)** and **stearic acid (18:0)**.

Unsaturated Fatty Acids: Contain one or more double bonds.

Monounsaturated (MUFAs): Have **one double bond**, such as **oleic acid** ($\text{C}_{18:1}$).

Polyunsaturated (PUFAs): Have **two or more double bonds**, such as **linoleic** and **arachidonic acid**. These double bonds are typically spaced at three-carbon intervals.

III. Isomerism and Membrane Fluidity

The physical state of a fatty acid is determined by its structure, which directly impacts the functionality of cell membranes.

Cis Configuration: In nature, double bonds are nearly always in the *cis* configuration. This creates a **"kink"** or bend in the hydrocarbon chain.

Membrane Fluidity: These kinks prevent tight packing of the fatty acid chains, **lowering the melting temperature (T_m)** and increasing the **fluidity** of the membrane.

Trans Configuration: *Trans* fatty acids, primarily produced through industrial **hydrogenation** of vegetable oils, lack these kinks and behave more like saturated fats. They elevate serum LDL cholesterol and are linked to an increased risk of **coronary heart disease**.

IV. Nomenclature and Essential Fatty Acids

Fatty acids are named based on their total carbon count and the position of their double bonds.

Numbering Systems: Carbons are numbered starting from the carboxyl carbon (Carbon 1). The methyl-terminal end is always called the **-carbon (or n-carbon)**.

-Nomenclature: This system counts from the methyl end until the first double bond is reached.

Essential Fatty Acids: Humans can introduce double bonds at carbons 4, 5, 6, and 9, but **lack the enzymes** (desaturases) to introduce double bonds past carbon 9 toward the -end. Therefore, two fatty acids are dietary essentials:

Linoleic acid (-6): The precursor for arachidonic acid, which is required for **prostaglandin** and eicosanoid synthesis.

α -Linolenic acid (-3): The precursor for EPA and DHA; these are vital for growth, brain development, and reducing cardiovascular risk.

Deficiency: Though rare, essential fatty acid deficiency results in **scaly dermatitis**, hair loss, and poor wound healing.

V. Biological Functions

Primary Fuel Source: Fatty acids are the body's **major energy reserve** when stored as triacylglycerols (TAG) in adipose tissue.

High Energy Yield: Because they are highly reduced and anhydrous, their complete oxidation yields approximately **9 kcal/g**, more than double that of carbohydrates or proteins. [24/09/2026 10:41 AM] Imaanaa Harmee: **Signaling:** They serve as precursors for **eicosanoids** (prostaglandins, thromboxanes, and leukotrienes), which act as local mediators of inflammation and other physiological processes.

Structural Components: They are integral to **phospholipids and glycolipids**, forming the hydrophobic core of all biological membranes.

Protein Anchoring: Fatty acids (like palmitate or myristate) can be covalently attached to proteins, helping them associate with **cell membranes**.

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Essential Fatty Acids

These are fatty acids that the **human body cannot synthesize de novo** and, therefore, must be obtained through the diet.

Linoleic acid (an -6 fatty acid): This 18-carbon polyunsaturated fatty acid is a precursor to **arachidonic acid**, which is the substrate for prostaglandin and eicosanoid synthesis.

α -Linolenic acid (an -3 fatty acid): This is a precursor for other long-chain -3 fatty acids,

such as **eicosapentaenoic acid (EPA)** and **docosahexaenoic acid (DHA)**, which are vital for growth, brain development, and cardiovascular health.

Why they are essential: Humans possess the enzymes (desaturases) required to introduce double bonds at carbons 4, 5, 6, and 9 of the fatty acid chain. However, we **lack the enzymes** necessary to introduce double bonds beyond carbon 9 toward the methyl () end of the chain. Because linoleic and linolenic acids have double bonds in these restricted positions, they are nutritional essentials provided by plants.

Clinical Significance of Deficiency: While rare, a deficiency in essential fatty acids can lead to:

Scaly dermatitis (ichthyosis).

Hair loss.

Poor wound healing.

Visual and neurologic abnormalities.

2. Non-essential Fatty Acids

Non-essential fatty acids are those that the body can synthesize from other dietary components.

Synthesis: The body can convert excess calories from **carbohydrates and proteins** into fatty acids through de novo synthesis in the cytosol, primarily in the liver.

Common Examples: Saturated fatty acids like **palmitic acid (16:0)** and **stearic acid (18:0)**, as well as monounsaturated fatty acids like **oleic acid ()**, can all be synthesized by human enzymes.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Essential fatty acids (EFAs), which include **linoleic acid** (an ω -6 fatty acid) and **α -linolenic acid** (an ω -3 fatty acid), are polyunsaturated fatty acids that humans cannot synthesize and must obtain from the diet. They serve several critical roles in human physiology:

1. Structural Components of Cell Membranes

EFAs are integral structural components of membrane lipids, such as **phospholipids and glycolipids**. Because they are highly unsaturated, they help maintain the **fluid nature** of biological membranes. This fluidity is essential for proper membrane function, including the movement of proteins and the transmission of signals across the cell surface.

2. Precursors for Signaling Molecules (Eicosanoids)

EFAs serve as the primary precursors for **eicosanoids**, a family of potent, hormone-like signaling molecules that act locally on neighboring cells.

ω -6 Pathway: Linoleic acid is desaturated and elongated to **arachidonic acid**, which is the immediate precursor for the predominant class of **prostaglandins, thromboxanes, and leukotrienes**. These molecules are major mediators of the inflammatory response.

ω -3 Pathway: -Linolenic acid is the precursor for other long-chain -3 fatty acids, such as **eicosapentaenoic acid (EPA)** and **docosahexaenoic acid (DHA)**, which also serve as precursors for signaling molecules with diverse physiological effects.

3. Growth and Development

EFAAs are vital for normal **growth and development**. Specifically, long-chain -3 polyunsaturated fatty acids (PUFAs) are included in infant formulas because they are critical for proper **brain development** and visual function.

4. Cardiovascular Health and Protection

The intake of -3 fatty acids, in particular, is associated with significant cardiovascular benefits:

Reducing Arrhythmias: They help suppress cardiac arrhythmias.

Lipid Lowering: They reduce levels of serum triacylglycerols.

Antithrombotic Effects: They decrease the tendency for thrombosis (blood clot formation) and platelet aggregation.

Blood Pressure: They have been shown to lower blood pressure and reduce the overall risk of cardiovascular mortality.

5. Additional Roles

Protein Anchoring: Fatty acids can be covalently attached to certain intracellular proteins, enhancing their ability to associate with cell membranes.

Skin Integrity: Specific long-chain fatty acids (derived from precursors) are major components of the skin and help regulate its water permeability. A deficiency in these essential fatty acids can lead to clinical symptoms such as **scaly dermatitis** (ichthyosis), hair loss, poor wound healing, and visual or neurologic abnormalities.

[24/09/2026 10:41 AM] Imaanaa Harmee:
ALCOHOLS ARE DERIVED OF LIPIDS

1. Glycerol: The Backbone of Simple and Complex Lipids

Glycerol is a three-carbon alcohol containing three hydroxyl groups.

▽ **Triacylglycerols (TAGs):** These simple lipids, which serve as the body's major energy reserve, consist of three fatty acids esterified to a single **glycerol** moiety. When TAGs are hydrolyzed (broken down) during the first stage of catabolism, they yield free fatty acids and **glycerol**.

▽ **Glycerophospholipids:** These complex lipids also use glycerol as a backbone, with fatty acids esterified at carbons 1 and 2, and a phosphate group at carbon 3.

2. Sphingosine: The Amino Alcohol Backbone

▽ **Sphingosine** is an amino alcohol derived from the amino acid serine and the fatty acid

palmitate.

It serves as the backbone for **sphingolipids**, such as sphingomyelin (found in the myelin sheath of nerve fibers) and glycolipids.

When a fatty acid is attached to the amino group of sphingosine, it forms a **ceramide**, which is the precursor for all sphingolipids.

3. Cholesterol: The Steroid Alcohol

▽ Cholesterol is specifically defined in the sources as the "characteristic **steroid alcohol** of animal tissues".

It is a very hydrophobic molecule containing a four-fused-ring steroid nucleus with a **hydroxyl (alcohol) group** at carbon 3.

As a sterol, it performs essential functions including modulating membrane fluidity and serving as a precursor for bile salts and steroid hormones.

4. Polar Head-Group Alcohols in Phospholipids

Phospholipids are composed of an alcohol attached by a phosphodiester bridge to either diacylglycerol or sphingosine. The sources identify several specific alcohols that serve as these "head groups":

▽ **Choline:** Forms phosphatidylcholine (lecithin).

▽ **Ethanolamine:** Forms phosphatidylethanolamine.

▽ **Serine:** Forms phosphatidylserine.

▽ **Inositol:** Forms phosphatidylinositol.

▽ **Glycerol:** Can also serve as a head group, forming phosphatidylglycerol or cardiolipin.

5. Other Lipid-Related Alcohols

▽ **Retinol (Vitamin A):** This fat-soluble vitamin is a primary alcohol containing a β -ionone ring.

▽ **β -Hydroxybutyrate:** One of the functional ketone bodies produced from fatty acid oxidation; it contains an **alcohol (hydroxyl) group** rather than a keto group, making the term "ketone body" technically a misnomer for this specific molecule.

Dolichol: A specialized lipid used in glycoprotein synthesis that is an isoprenoid alcohol.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Triacylglycerols (TAGs)

▽ also known as **triglycerides**, are the primary form of **energy storage** in the human body.

They are characterized by their extreme hydrophobicity and are highly efficient as a fuel source due to their reduced state and anhydrous storage.

I. Structure and Composition

▽ **Chemical Definition:** A triacylglycerol consists of **three fatty acids esterified to a single glycerol moiety**.

▽ **"Neutral Fat":** Because the negative charges of the fatty acid carboxyl groups are lost through esterification, triacylglycerols are often referred to as **neutral fats**.

▽ **Mixed Triacylglycerols:** TAGs rarely contain the same fatty acid at all three positions. Typically, carbon 1 contains a **saturated** fatty acid, carbon 2 contains an **unsaturated** fatty acid (which lowers the melting temperature), and carbon 3 can be either.

II. Role as Energy Reserve

▽ **Energy Density:** Triacylglycerols are highly reduced and yield approximately **9 kcal/g** when completely oxidized—more than twice the energy provided by an equivalent amount of carbohydrate or protein.

▽ **Anhydrous Storage:** Unlike glycogen, which is polar and binds approximately four times its weight in water, triacylglycerols are nonpolar and coalesce as **anhydrous (water-free) droplets** in the cytosol of adipose cells. This allows the body to store vast amounts of energy with minimal added weight.

▽ **Major Energy Reserve:** They serve as the body's **major energy reserve**, primarily stored in white adipose tissue.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Glycerophospholipids

✓ Also known as **phosphoglycerides** or **phosphoacylglycerols** are the major class of phospholipids and serve as the primary structural components of biological membranes.

1. Structure and Composition

All glycerophospholipids are built upon a **glycerol backbone** and contain (or are derived from) **phosphatidic acid (PA)**.

◆ **Phosphatidic Acid:** This is the simplest phosphoglyceride, consisting of a glycerol molecule with fatty acids esterified at carbons 1 and 2, and a **phosphate group** attached to carbon 3.

◆ **Amphipathic Nature:** They possess a **hydrophilic (polar) head** (the phosphate group plus an attached alcohol) and a **long hydrophobic (nonpolar) tail** (the fatty acid chains).

♦**Fatty Acid Distribution:** Typically, carbon 1 contains a **saturated** fatty acid, while carbon 2 contains an **unsaturated** fatty acid (such as arachidonic acid).

2. Major Classes of Glycerophospholipids

These are named based on the alcohol group esterified to the phosphate on the PA backbone:

♦**Phosphatidylcholine (Lecithin):** One of the most abundant phospholipids in cell membranes and a key component of lung surfactant.

♦**Phosphatidylethanolamine (Cephalin) and Phosphatidylserine:** Principal membrane phospholipids; phosphatidylserine is specifically found in the **inner leaflet** of the plasma membrane, contributing to its negative charge.

♦**Phosphatidylinositol (PI):** Acts as a reservoir for **arachidonic acid** and a precursor for second messengers.

♦**Cardiolipin (Diphosphatidylglycerol):** Composed of two PA molecules linked by a glycerol; it is virtually exclusive to the **inner mitochondrial membrane**, where it supports the electron transport chain.

Ether Lipids:

♦**Plasmalogens:** Contain an unsaturated alkyl group attached by an **ether linkage** at carbon 1 (e.g., phosphatidylethanolamine in nerve tissue).

♦**Platelet-Activating Factor (PAF):** A potent mediator of inflammation and protein-cell interaction that triggers thrombotic events.

3. Biological Functions

◇**Membrane Integrity:** They form the **lipid bilayer**, providing a hydrophobic barrier that partitions aqueous cellular compartments.

◇**Intracellular Signaling:** Membrane-bound **phosphatidylinositol 4,5-bisphosphate (PIP₂)** is cleaved by **phospholipase C** to generate the second messengers **IP₃** and **diacylglycerol (DAG)**.

◇**Protein Anchoring:** Specific proteins are covalently attached to the external membrane surface via **glycosyl phosphatidylinositol (GPI)** glycan anchors.

◇**Specialized Roles:** **Dipalmitoylphosphatidylcholine (DPPC)** is the major component of **lung surfactant**, which reduces surface tension to prevent alveolar collapse.

4. Synthesis and Degradation

▽ **Synthesis:** Occurs primarily in the **smooth endoplasmic reticulum (SER)**. The process requires the activation of either the diacylglycerol or the alcohol head group using **cytidine**

diphosphate (CDP).

▽ **Degradation:** Performed by **phospholipases (A1, A2, C, and D)**, which hydrolyze specific phosphodiester bonds.

▽ **Phospholipase A2:** Releases arachidonic acid for prostaglandin synthesis and is inhibited by **glucocorticoids**.

Clinical Relevance: Insufficient production of DPPC results in **Respiratory Distress Syndrome (RDS)** in preterm infants. Bacterial toxins, such as those from *Clostridium perfringens*, use phospholipases to dissolve host cell membranes.

[24/09/2026 10:41 AM] Imaanaa Harmee:

II. Primary Classes of Glycerophospholipids

These classes are named based on the specific alcohol head group attached to the phosphatidic acid backbone:

Phosphatidylcholine (Lecithin):

Composition: Formed by adding **choline** to PA.

Significance: It is the most abundant phospholipid in the cell membranes of most eukaryotic cells.

Specialized Role: **Dipalmitoylphosphatidylcholine (DPPC)** is the major lipid component of **lung surfactant**, which reduces surface tension in the alveoli to prevent lung collapse.

Phosphatidylethanolamine (Cephalin):

Composition: Formed by adding **ethanolamine** to PA.

Significance: Like PC, it is one of the most abundant phospholipids in eukaryotic cells. It is primarily concentrated in the **inner leaflet** of the plasma membrane.

Phosphatidylserine (PS):

Composition: Formed by adding **serine** to PA.

Significance: It is found primarily in the **inner leaflet** of the plasma membrane. At physiologic pH, it carries a **net negative charge**, which contributes to the membrane potential and aids in binding positively charged molecules within the cell.

Phosphatidylinositol (PI):

Composition: Formed by adding **inositol** to PA.

Significance: PI is unusual because it frequently contains **stearic acid** on carbon 1 and **arachidonic acid** on carbon

2. It serves two vital specialized roles:

Reservoir for Arachidonic Acid: It provides the necessary substrate for **prostaglandin synthesis**.

Intracellular Signaling: PI is phosphorylated to form **PIP(2)**, which can be cleaved into the second messengers **IP(3)** and **DAG** to mediate cellular responses to hormones.

Protein Anchoring: Specific proteins (e.g., alkaline phosphatase) are attached to cell surfaces via **glycosyl phosphatidylinositol (GPI) anchors**.

III. Specialized Mitochondrial and Precursor Lipids

Phosphatidylglycerol (PG): Occurs in large amounts in mitochondrial membranes and serves as the immediate precursor for the synthesis of cardiolipin.

Cardiolipin (Diphosphatidylglycerol): This molecule consists of two PA molecules linked by an additional glycerol. In eukaryotes, cardiolipin is **virtually exclusive to the inner mitochondrial membrane**, where it is essential for maintaining the respiratory complexes of the **electron transport chain**.

IV. Ether Glycerophospholipids

These differ from standard glycerophospholipids because they contain an **ether linkage** at carbon 1 rather than an ester linkage.

Plasmalogens: These contain an unsaturated alkyl group.

Phosphatidylethanolamine is abundant in nerve tissue.

Phosphatidylcholine is abundant in heart muscle.

Platelet-Activating Factor (PAF): This is a unique ether lipid with a saturated alkyl group at carbon 1 and an **acetyl residue** (instead of a fatty acid) at carbon 2. PAF is a potent bioactive molecule that triggers **thrombotic and inflammatory events** even at extremely low concentrations.

[24/09/2026 10:41 AM] Imaanaa Harmee:

Sphingolipids

✔ are a complex class of lipids that, unlike glycerolipids, do not have a glycerol backbone; instead, they are built upon the long-chain amino alcohol **sphingosine**. These molecules are essential structural components of cell membranes and are particularly abundant in **nerve tissue**.

I. Structural Foundation: Ceramide

The fundamental building block of all sphingolipids is **ceramide**.

▽ **Composition:** A ceramide consists of a sphingosine molecule with a long-chain fatty acid attached to its amino group via an **amide linkage**.

▽ **Precursor Role:** Ceramide serves as the immediate precursor for both phosphorylated and glycosylated sphingolipids.

II. Major Classes of Sphingolipids

Sphingolipids are categorized based on the chemical group attached to the hydroxyl group of the ceramide backbone:

Sphingomyelin (Sphingophospholipids):

✓ **Structure:** Formed when the alcohol group at carbon 1 of sphingosine is esterified to **phosphorylcholine**.

✓ **Function:** It is the only significant sphingophospholipid in humans and is a vital component of the **myelin sheath**, which insulates and protects nerve fibers.

Glycosphingolipids (Glycolipids): These contain carbohydrate moieties attached directly to the ceramide by an O-glycosidic bond and do not contain phosphate.

Neutral Glycosphingolipids:

✓ **Cerebrosides:** The simplest neutral glycolipids, containing a single sugar residue (typically **galactose** in myelin or **glucose** in other tissues).

✓ **Globosides:** Produced by attaching additional neutral monosaccharides to a glucocerebroside.

Acidic Glycosphingolipids (Negatively charged at physiologic pH):

✓ **Gangliosides:** The most complex sphingolipids, containing one or more molecules of **N-acetylneuraminic acid (NANA)**. They are primarily found in ganglion cells of the central nervous system.

✓ **Sulfatides:** Cerebrosides that contain a **sulfated galactosyl residue**.

III. Synthesis and Degradation

✓ **Synthesis:** Sphingosine is synthesized from **palmitoyl CoA and serine** in a reaction requiring pyridoxal phosphate (Vitamin B6). Further synthesis of glycosphingolipids occurs primarily in the **Golgi apparatus** through the sequential addition of sugars from UDP-sugar donors.

✓ **Degradation:** Sphingolipids are internalized by endocytosis and degraded within **lysosomes** by specific acid hydrolases. This follows a sequential "last on, first off" rule, where the last group added during synthesis is the first removed.

IV. Clinical Significance: Sphingolipidoses

Hereditary deficiencies in the lysosomal enzymes required to degrade sphingolipids lead to **sphingolipidoses**, characterized by the abnormal accumulation of specific lipids in tissues.

♦ **Niemann-Pick Disease (Types A and B):** Caused by a deficiency in **sphingomyelinase**, leading to sphingomyelin accumulation in the liver and spleen, and severe neurodegeneration in the infantile form (Type A).

♦**Tay-Sachs Disease:** A deficiency in **hexosaminidase A** results in the accumulation of **GM2 gangliosides**, leading to rapid, fatal neurodegeneration, blindness, and a characteristic cherry-red macula.

♦**Gaucher Disease:** The most common lysosomal storage disease, caused by a deficiency of **β -glucosidase**, resulting in glucocerebroside accumulation. Symptoms include hepatosplenomegaly and osteoporosis.

♦**Fabry Disease:** An X-linked disorder caused by **α -galactosidase** deficiency, leading to globoside accumulation and kidney/heart failure

[24/09/2026 10:42 AM] Imaanaa Harmee:

Glycolipids

✔ more precisely called **glycosphingolipids**, are molecules containing both carbohydrate and lipid components that are essential structural constituents of all cell membranes. They are found in greatest amounts in **nerve tissue**, particularly within the **myelin sheath**.

I. Structure and Composition

✧ Glycolipids are derivatives of **ceramides**, which consist of the amino alcohol **sphingosine** with a long-chain fatty acid attached via an amide linkage.

🔗 **Linkage:** Unlike sphingomyelin (a phospholipid), glycolipids **do not contain phosphate**. Instead, their polar head is a monosaccharide or oligosaccharide attached directly to the ceramide by an **O-glycosidic bond**.

📍 **Location:** They are primarily located in the **outer leaflet** of the plasma membrane, where the carbohydrate portion extends into the extracellular space to form part of the **glycocalyx**.

II. Classification

✧ Glycolipids are categorized based on their carbohydrate moieties and charge:

A. Neutral Glycosphingolipids

These are uncharged at physiologic pH.

△ **Cerebrosides:** The simplest neutral glycolipids, containing a single sugar molecule—typically **galactose** (galactocerebroside, common in myelin) or **glucose** (glucocerebroside, an intermediate in complex lipid synthesis).

△ **Globosides:** Produced by attaching additional neutral monosaccharides to a glucocerebroside.

B. Acidic Glycosphingolipids

These carry a negative charge at physiologic pH.

△ **Gangliosides:** The most complex glycosphingolipids, found primarily in CNS ganglion

cells. They contain one or more molecules of **N-acetylneuraminic acid (NANA)**, which provides the negative charge.

△ **Sulfatides:** Cerebrosides containing **sulfated galactosyl residues**. The sulfate group is donated by **PAPS** (3'-phosphoadenosine-5'-phosphosulfate).

III. Biological Functions

Glycolipids play critical roles in regulating cellular interactions and protecting the cell:

△ **Cell Surface Recognition:** They act as markers for cell-to-cell signaling and adhesion.

△ **Antigenicity:** They are a source of **blood group antigens**, embryonic antigens, and tumor-specific antigens.

△ **Receptors:** They serve as cell surface receptors for various **bacterial toxins** (e.g., cholera and tetanus toxins), as well as certain viruses and microbes.

IV. Synthesis and Degradation

☉ **Synthesis:** Occurs primarily in the **Golgi apparatus** through the sequential addition of sugars from **UDP-sugar donors** (like UDP-glucose or UDP-galactose).

☉ **Degradation:** Glycolipids are internalized by endocytosis and degraded in **lysosomes** by specific **acid hydrolases**. This follows a "last on, first off" sequential process.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Revision

☞ The nomenclature of fatty acids involves two primary systems for identifying carbon atoms in the chain, along with a shorthand notation for describing the length and degree of unsaturation.

1. Standard Numbering System (Carboxyl End)

In this system, carbon atoms are numbered beginning with the **carboxyl carbon as Carbon 1**.

☞ **Greek Letter Designations:** Carbon 2 (the carbon next to the carboxyl group) is called the **α-carbon**, Carbon 3 is the **β-carbon**, and Carbon 4 is the **γ-carbon**.

☞ **Δ (Delta) Notation:** The positions of double bonds are designated relative to the carboxyl end using the Δ symbol. For example, **18:1, Δ9** describes a fatty acid with 18 carbons and one double bond between carbons 9 and 10. This is also commonly written as **18:1(9)**.

2. ω (Omega) or "n" Numbering System (Methyl End)

This system identifies the position of double bonds relative to the **methyl-terminal end** of the chain.

☞ **ω-Carbon:** The carbon of the terminal methyl group is always called the **ω-carbon** (or n-carbon), regardless of the chain length.

🔗 **ω -Series:** Fatty acids are classified by the distance of the first double bond from the ω end. For instance, **arachidonic acid** (20:4, Δ 5,8,11,14) is an ω -6 (or n-6) fatty acid because its terminal double bond is six carbons away from the ω end. Similarly, **α -linolenic acid** is an ω -3 fatty acid.

3. Shorthand Notation

A standard shorthand is used to quickly convey the structure of a fatty acid:

🔗 **Number before the colon:** Indicates the total number of carbon atoms in the chain.

🔗 **Number after the colon:** Indicates the total number of double bonds.

🔗 **Numbers in parentheses/superscript:** Indicate the specific positions of those double bonds.

Example: Stearic acid is 18:0 (18 carbons, 0 double bonds).

Example: Arachidonic acid is 20:4(5,8,11,14) (20 carbons, 4 double bonds at the specified positions).

4. Common vs. Systematic Names

🔗 **Saturated Fatty Acids:** Most have common names based on their origins, such as **palmitic acid** (16 carbons) and **stearic acid** (18 carbons). Shorter chains may use Latin-derived names like **octanoic acid** (8 carbons) or **decanoic acid** (10 carbons).

🔗 **Unsaturated Configurations:** Double bonds in naturally occurring fatty acids are nearly always in the *cis* configuration, which causes the chain to "kink". Fatty acids produced by industrial hydrogenation often contain *trans* double bonds.

[24/09/2026 10:46 AM] Imaanaa Harmee:
Protein

[24/09/2026 10:46 AM] Imaanaa Harmee:
Proteins

▽ the most abundant and functionally diverse molecules in living systems, participating in virtually every life process. The term "protein" originates from the Greek word *proteios*, meaning "first place," a name suggested by the Swedish chemist Berzelius to signify their primary importance to life.

I. Composition and Structural Conventions

▽ **Building Blocks:** Proteins are linear polymers composed of **20 different L--amino acids**.

▽ **The Peptide Bond:** These amino acids are joined covalently by **peptide bonds**, which are amide linkages between the -carboxyl group of one amino acid and the -amino group of the next.

▽ **Chain Lengths:** While the majority of polypeptide chains have fewer than 2,000 amino acids, lengths vary significantly. For instance, **human cytochrome c** contains 104 residues, whereas the muscle protein **titin** is exceptionally large at approximately 27,000 residues.

▽ **Nomenclature and Reading Direction:** By convention, the sequence of amino acids is always read from the **N-terminal end** (the free amino group on the left) to the **C-terminal end** (the free carboxyl group on the right).

II. Protein Organization

Proteins are classified based on the number and type of polypeptide chains (subunits) they contain:

▽ **Monomeric Proteins:** Consist of a single polypeptide chain.

▽ **Multimeric/Oligomeric Proteins:** Consist of two or more polypeptide chains (subunits).
Homomultimeric: Identical subunits.

Heteromultimeric: Different types of subunits.

▽ **Protomers:** In proteins with nonidentical subunits, the unit structure composed of these nonidentical parts is called a protomer. For example, **hemoglobin** is a tetramer (4) that can be viewed as being composed of two identical -protomers.

III. Diverse Biological Functions

The functional diversity of proteins is a result of their unique three-dimensional folding dictated by their primary amino acid sequence.

▽ **Catalysts:** Virtually all chemical reactions in the body are mediated by **enzymes**, which are protein catalysts that increase reaction rates.

▽ **Mechanical Work: Contractile proteins** in muscle (like **actin and myosin**) permit movement, while tubulin-based structures like **cilia and flagella** facilitate cellular locomotion.

▽ **Cargo Transport:** Motor proteins like **kinesins** (moving toward the "plus" end) and **dyneins** (moving toward the "minus" end) use ATP energy to transport molecules and organelles along microtubules.

▽ **Transport of Molecules:** Proteins such as **hemoglobin** transport oxygen in the blood, **transferrin** transports iron, and **albumin** carries various hydrophobic molecules.

▽ **Storage (Reservoir) Proteins:** **Ferritin** stores iron, **myoglobin** stores oxygen in muscle, and proteins like **casein** (in milk) or **albumin** (in egg whites) serve as nutrient reservoirs.

▽ **Hormones and Regulation:** Polypeptide hormones like **insulin** and **growth hormone (GH)** direct and regulate body metabolism.

▼ **Structural Components:** **Collagen** provides a framework for bone and connective tissue, acting like steel cables in reinforced concrete. **Keratin** is a major component of hair and wool.

▼ **Protection:** **Immunoglobulins** (antibodies) fight infectious agents, while **clotting factors** (like thrombin) prevent blood loss.

▼ **Exploitive/Toxic Proteins:** Certain organisms produce proteins for offense or defense, such as snake venoms or bacterial toxins (e.g., **cholera toxin**).

▼ **Pathogenic Misfolding (Prions):** Diseases like "Mad Cow Disease" (TSEs) are caused by **prions**, altered versions of normal proteins that act as templates to cause other proteins to misfold into neurotoxic assemblies.

▼ **Self-Defense and Luminescence:** Some organisms use proteins for unique defensive or signaling mechanisms, such as the **Green Fluorescent Protein (GFP)** in jellyfish. In fireflies, the protein **luciferin** reacts with ATP in a process catalyzed by **luciferase** to produce light.
[24/09/2026 10:46 AM] Imaanaa Harmee: Proteins can be classified based on their structural shape, the complexity of their organization, their subunit composition, and their biological functions.

I. Major Structural Classifications

Proteins are generally grouped into three broad categories based on their overall three-dimensional shape and solubility:

Globular Proteins: These resemble irregular balls and are usually soluble in aqueous media. They are characterized by a densely packed hydrophobic core with polar side chains on the surface. Examples include **myoglobin**, **hemoglobin**, and most **enzymes**.

Fibrous Proteins: Geometrically linear and arranged around a single axis, these proteins have a repeating unit structure and serve primarily structural functions. They are often insoluble and provide mechanical support. Examples include **collagen** (the most abundant protein in the body) and **keratin**.

Transmembrane Proteins: These consist of proteins with one or more regions (typically - helices with hydrophobic side chains) that span the lipid bilayer of cell membranes. Examples include **ion channels** and **G-protein-coupled receptors**.

II. Classification by Hierarchical Levels

The complexity of protein architecture is organized into four levels:

Primary Structure: The unique linear sequence of amino acids in a polypeptide chain.

Secondary Structure: Localized regular arrangements of the polypeptide backbone, such as **-helices**, **-sheets**, and **-bends** (turns), stabilized by hydrogen bonds.

Tertiary Structure: The overall three-dimensional folding of the secondary structural elements and domains into a compact native conformation.

Quaternary Structure: The arrangement and association of two or more polypeptide subunits (e.g., the tetrameric structure of hemoglobin).

III. Classification by Composition

Proteins can be categorized by whether they consist solely of amino acids or include non-protein components:

Simple Proteins: Composed only of amino acid residues.

Conjugated (Complex) Proteins: These contain a tightly bound non-protein **prosthetic group**.

Hemoproteins: Contain heme as a prosthetic group (e.g., hemoglobin, myoglobin, and cytochromes).

Glycoproteins: Proteins with attached branched heterooligosaccharide chains, often found on cell surfaces for recognition and antigenicity.

Proteoglycans: Consist of a core protein covalently attached to long, unbranched glycosaminoglycan chains, resembling a "bottle brush".

IV. Classification by Subunit Organization

Proteins are also classified by the number and type of polypeptide chains they contain:

Monomeric Proteins: Consist of a single polypeptide chain.

Multimeric (Oligomeric) Proteins: Consist of two or more subunits.

Homomultimeric: Identical subunits (e.g., CK-MM or CK-BB isozymes).

Heteromultimeric: Different subunits (e.g., hemoglobin, which is a tetramer of two and two chains).

Protomers: The unit structure composed of nonidentical subunits in a multimeric protein (e.g., an α -pair in hemoglobin).

V. Functional Classification

Proteins are classified by the physiological roles they perform in the body:

Catalysts (Enzymes): Increase the rate of chemical reactions (e.g., lactate dehydrogenase).

Transport Proteins: Shuttle essential molecules (e.g., **hemoglobin** for oxygen, **albumin** for hydrophobic compounds).

Structural Proteins: Provide frameworks for tissues (e.g., **collagen** in bone, **elastin** in blood).

vessels).

Hormones and Signaling: Direct and regulate metabolism (e.g., **insulin**, **growth hormone**).

Protection: Participate in immune responses (e.g., **immunoglobulins**) or blood clotting (e.g., **thrombin**).

Contractile/Motor Proteins: Permit movement and cargo transport (e.g., **actin** and **myosin** in muscle).

VI. Specialized Classifications

Isoforms/Isozymes: Proteins that perform the same function but have different primary structures (e.g., the different tissue-specific forms of **creatine kinase**).

Zymogens (Proenzymes): Inactive precursors that require proteolytic cleavage to become active enzymes (e.g., **trypsinogen** or **proinsulin**).

[24/09/2026 10:46 AM] Imaanaa Harmee: **Chaperones:** Specialized proteins (heat-shock proteins) required for the proper folding of other proteins.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Simple proteins

defined as molecules composed exclusively of amino acid residues that yield only amino acids upon hydrolysis. They represent the majority of **plasma proteins**, which maintain a total concentration of **65–80 g/L** and are primarily synthesized in the liver, with specific contributions from enterocytes and lymphocytes.

I. Plasma Proteins

Plasma proteins are categorized into **albumin** and various **globulins**.

Albumin (35–50 g/L):

♦**Functions:** Its primary role is the maintenance of **plasma oncotic (colloidal-osmotic) pressure**, which keeps water within the blood vessels. It also serves as a critical transport protein for hydrophobic substances, including **bilirubin**, **free fatty acids**, and **steroid hormones** (such as testosterone and aldosterone).

♦**Clinical Significance:** A decrease in albumin levels (hypoalbuminemia, <20 g/L) leads to water entering the interstitial space, resulting in **edema**. This is a hallmark of **Kwashiorkor**, a form of protein-energy malnutrition.

Globulins (20–35 g/L): These are classified by weight and electrophoretic mobility into α , β , and γ types.

▽ α_1 -**Antitrypsin** (α_1 -**AT**): A serine protease inhibitor synthesized by liver cells and macrophages. It prevents the degradation of **elastin** in the lungs by inhibiting neutrophil elastase. Deficiency in α_1 -AT leads to **emphysema**.

▽ **Haptoglobin** (α_2): (As noted in the provided outline) This protein conserves

hemoglobin and iron by binding free Hb, preventing its loss through the kidneys and avoiding tubular damage.

▽ **Transferrin (β):** A liver-synthesized protein that binds and transports free iron. By sequestering iron, it prevents it from participating in the Fenton reaction, which produces toxic hydroxyl radicals; thus, it acts as an antioxidant.

▽ **γ -Globulins (Immunoglobulins):** These are antibodies (IgG, IgA, IgM, IgD, IgE) produced by B-lymphocytes/plasma cells to defend against foreign antigens.

II. Globins (Histones)

These are basic proteins rich in **Histidine** and **Lysine** residues.

● **Histones:** Found in chromosomes associated with DNA, where they can undergo posttranslational modifications like **acetylation**.

● **Globins:** These combine with **heme** (a prosthetic group) to form hemoproteins such as **myoglobin** in muscle and **hemoglobin** in red blood cells.

III. Scleroproteins (Fibrous Proteins)

These are indigestible, structurally essential "animal skeleton proteins."

✧ **α -Keratin:** A major fibrous component of hair, nails, and skin, characterized by a nearly entirely **α -helical structure**.

✧ **Collagens:** The most abundant proteins in the body. They form a **triple-stranded helix** and provide structural integrity to bones, teeth, tendons, and skin.

✧ **Elastin:** A connective tissue protein with **rubber-like properties**, allowing it to stretch and recoil. It is found in high concentrations in the lungs and the walls of large blood vessels like the **aorta**.

IV. Gliadins

Gliadins are **cereal proteins** found in gluten. In patients with **Celiac disease**, the ingestion of gliadin causes immune-mediated damage to the small intestine, leading to malabsorption.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Conjugated proteins

✧ are complex molecules consisting of a protein part (the **apoprotein**) and a non-protein part (the **prosthetic group**). When the protein is combined with its prosthetic group, the functional unit is called a **holoprotein**. These proteins are classified based on the chemical nature of their non-protein component.

1. Phosphoproteins

☞ **Phosphoproteins** are proteins where a **phosphate group** is covalently attached to the hydroxyl (-OH) group of specific amino acids.

☞ **Mechanism:** This modification is catalyzed by a family of enzymes called **protein kinases**, which use ATP as the phosphate donor. This process is reversible through the action of **phosphoprotein phosphatases**.

☞ **Target Residues:** Phosphate is typically attached to **serine**, **threonine**, or, less frequently, **tyrosine** residues.

Examples: Notable examples include **casein**, the primary protein in milk, and **ovovitellin**, found in egg yolk.

2. Lipoproteins

Lipoproteins are spherical macromolecular complexes of lipids and specific proteins called **apolipoproteins**.

🔗 **Structure:** They consist of a neutral lipid core (triacylglycerol and cholesteryl esters) surrounded by a shell of **amphipathic apolipoproteins**, phospholipids, and unesterified cholesterol.

🔗 **Function:** They keep hydrophobic lipids soluble while transporting them through the aqueous environment of the plasma.

🔗 **Major Classes:** They are classified by density into **chylomicrons**, **very-low-density lipoproteins (VLDL)**, **low-density lipoproteins (LDL)**, and **high-density lipoproteins (HDL)**.

3. Glycoproteins and Mucoproteins

These consist of proteins with **oligosaccharide chains** covalently attached to their side chains.

☞ **Glycoproteins:** Typically contain relatively short (2–10 sugar residues), often branched carbohydrate chains. Examples include **immunoglobulin IgG** (containing <4% carbohydrate) and **erythropoietin** [52, User Query].

☞ **Mucoproteins (Mucins/Proteoglycans):** These contain a much higher percentage of carbohydrate, often exceeding 80% of their mass, such as **gastric mucin**. **Proteoglycans** resemble a "bottle brush," with long, unbranched glycosaminoglycan chains radiating from a core protein.

☞ **Linkages:** The carbohydrate may be **N-linked** to the amide nitrogen of asparagine or **O-linked** to the hydroxyl group of serine or threonine.

☉ **Functions:** They are essential for **cell surface recognition**, antigenicity (like blood group antigens), and biological lubrication.

4. Nucleoproteins

Nucleoproteins are complexes where basic proteins are conjugated with nucleic acids (**DNA or RNA**).

🦋 **Histones:** These are small, basic proteins rich in **arginine and lysine**. Because they are positively charged at physiologic pH, they form strong ionic bonds with negatively charged DNA to form **nucleosomes**.

Examples:

✓ **Chromosomes:** Formed by the complex of DNA and histones (known as **chromatin**).

✓ **Ribosomes:** Composed of ribosomal RNA (rRNA) and proteins; they serve as the site for protein synthesis.

5. Metalloproteins

Metalloproteins contain tightly bound **metal ions** that are required for their biological or catalytic activity.

a. Iron-containing proteins:

✓ **Hemeproteins:** Contain heme (protoporphyrin IX and Fe^{2+}) as a prosthetic group. Examples include **hemoglobin** and **myoglobin** (oxygen transport), **cytochromes** (electron transport), and **catalase**.

✓ **Transport and Storage:** **Transferrin** transports iron in the plasma, while **ferritin** is the primary cellular storage protein for iron.

b. Copper-containing proteins: These include enzymes like **cytochrome c oxidase** and **lysyl oxidase** (involved in collagen cross-linking). **Ceruloplasmin** is a major copper carrier, transporting approximately 90% of serum copper.

c. Zinc-containing proteins: Zinc acts as an electrophile to assist in catalysis. Examples include **carbonic anhydrase** and **alcohol dehydrogenase**.

d. Magnesium-containing proteins: Mg^{2+} is often involved in binding negatively charged phosphate groups, making it a critical cofactor for **kinases** (which use ATP) and **phosphatases**.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Protein solubility is primarily determined by a protein's three-dimensional shape and the distribution of its amino acid side chains. Proteins are generally categorized into two main groups based on their structural shape and solubility in aqueous media: **globular proteins** and **fibrous proteins**.

I. Non-Water Soluble Proteins: Fibrous Proteins

✦ Fibrous proteins are generally **insoluble in aqueous medium**. Their structure is geometrically linear, arranged around a single axis, and typically consists of repeating unit structures.

✦ **Structural Characteristics:** They are often composed of specific amino acids combined into regular, secondary structural elements, providing them with unique mechanical properties. They frequently feature extensive interchain covalent or hydrogen bonding to increase tensile strength and stability.

Examples:

♦ **Keratins:** A family of fibrous proteins whose structure is nearly entirely α -helical; they are major components of hair and skin.

♦ **Collagen and Elastin:** Key structural components of the extracellular matrix, connective tissue, and blood vessel walls.

♦ **Contractile Proteins:** While the user query mentions **myosin** and **actin**, the sources note that G-actin (globular actin) polymerizes to form F-actin (fibrous actin), which makes up the microfilaments of the cytoskeleton.

II. Water-Soluble Proteins: Globular Proteins

✦ Globular proteins resemble irregular balls and are **usually soluble in aqueous medium**. This group includes most enzymes and transport proteins.

♦ **Structural Basis for Solubility:** When globular proteins fold into their native conformation, they generally form a **densely packed hydrophobic core**, burying nonpolar side chains in the interior. The **polar and charged amino acid side chains are located on the surface**, where they interact with the aqueous solvent through hydrogen bonds and ionic interactions, which enhances solubility.

♦ **Denaturation:** Solubility can be lost through **denaturation**, where heat, pH changes, or solvents disrupt the ionic, hydrogen, and hydrophobic bonds holding the structure together, causing the protein to unfold and often precipitate from solution.

III. Specific Soluble Protein Classes

✦ **Albumins:** These are readily soluble in water. **Serum albumin** is a major plasma protein that functions as a transporter for various molecules. Heat causes albumin to denature and convert from a translucent state into an insoluble **white precipitate** (coagulation).

✦ **Globulins:** These proteins have a higher molecular weight than albumin and are classified into α , β , and γ types. They are characterized by being soluble in dilute salt solutions but remaining sparingly soluble in pure water.

✦ **Histones:** These are small, water-soluble proteins that are **positively charged at physiologic pH** due to a high content of lysine and arginine. This strong positive charge allows them to form ionic bonds with the negatively charged phosphate groups of **nucleic acids (DNA)**, facilitating the packing of DNA into nucleosomes.

[24/09/2026 10:46 AM] Imaanaa Harmee:

fibrous proteins

(also known as fibrillar proteins or scleroproteins) are characterized by their elongated, linear shapes and their essential roles in providing structural integrity to the body.

I. Overall Shape and Structure

Geometrically Linear: Unlike globular proteins, which are compact and spherical, fibrous proteins are **geometrically linear** and typically arranged around a **single axis**.

Elongated Polypeptide Chains: They consist of elongated polypeptide chains that form a **repeating unit structure**.

Secondary Structure Dominance: Their special mechanical properties result from combining specific amino acids into **regular, secondary structural elements**. For example, the structure of keratins is nearly entirely α -helical.

Axial Ratio: they define fibrous proteins by their **elongated** nature. (Note: In biochemistry, the "axial ratio" refers to the ratio of a molecule's length to its width; for fibrous proteins, this ratio is characteristically high, typically greater than 10).

II. Biological Roles

Fibrous proteins are primarily found in the **extracellular matrix** and serve several critical "animal skeleton" functions:

Protective: They form the outer layers of the skin and components of hair and nails (e.g., **keratin**).

Connective and Supportive: They provide a framework for tissues. For instance, **collagen** serves as a semipermeable filtration barrier in basement membranes and provides high tensile strength in tendons and the cornea.

Elasticity: They allow tissues to stretch and recoil, which is vital for the lungs and large blood vessel walls (e.g., **elastin**).

III. Key Examples

Collagens: The most abundant proteins in the human body. They consist of three α chains wound into a **rope-like triple helix**. They provide great tensile strength to bones, teeth, and cartilage.

Elastin: A connective tissue protein with **rubber-like properties**. It forms an interconnected, rubbery network that can stretch to several times its normal length and recoil to its original shape.

Keratins: Closely related fibrous proteins that are a major component of hair and skin. Their rigidity is determined by the number of **disulfide bonds** between the constituent polypeptide

chains.

α 1-Antitrypsin (α 1-AT) Connection: Although α 1-AT is a globulin, its primary role is to **prevent the degradation of elastin** by inhibiting neutrophil elastase in the alveolar walls of the lungs.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Keratins

△ are a family of mechanically durable fibrous proteins that serve as the primary structural components of the vertebrate "skeleton" system, including hair, wool, nails, claws, and the outer layer of skin. They are classified as **intermediate filaments**, which provide structural support and resilience to the cell.

I. Structural Classification and Composition

△ Keratins are distinguished by their secondary structure and biological distribution:

△ **α -Keratins (Mammalian):** These are found exclusively in mammals and consist of polypeptide chains that are nearly entirely **α -helical**.

△ **Helix Nature:** The α -keratin helix is a right-handed spiral stabilized by intrachain hydrogen bonds.

△ **Amino Acid Profile:** They are rich in hydrophobic amino acids such as **Alanine, Valine, Leucine, Isoleucine, Methionine, and Phenylalanine** (AVLIMF), which cluster to exclude water.

△ **Cysteine and Strength:** α -keratins are notably rich in **Cysteine** residues. The rigidity and mechanical strength of the tissue—whether it is soft skin or a hard rhinoceros horn—is determined by the number of **disulfide bonds** cross-linking the constituent polypeptide chains.

△ **β -Keratins (Avian and Reptilian):** These are found in birds and reptiles (e.g., scales, feathers) and are also present in silk and spider webs. Unlike α -keratins, they are rich in small amino acids like **Glycine and Alanine** and are relatively deficient in Cysteine.

II. Molecular Organization

△ The formation of keratin filaments follows a hierarchical assembly process:

△ **Monomer:** An α -helical core with globular domains at the ends.

△ **Coiled-coil:** Two helical segments coil around each other to form a dimer.

△ **Protofilaments and Filaments:** These dimers combine into tetramers, which then join end-to-end to form protofilaments. Approximately eight protofilaments combine to form the final, thick **intermediate filament** (cytokeratin).

△ **Macro-structure:** Disulfide bridges occur between these dimers and other proteins within microfibrils and macrofibrils to provide extreme durability.

III. Clinical Significance: Keratinopathies

△ Genetic mutations in keratin genes lead to diseases characterized by the loss of mechanical integrity in epithelial tissues, often manifesting as **skin blistering**. This blistering reflects the skin's diminished capacity to resist mechanical stress due to structural abnormalities.

○ **Epidermolysis Bullosa Simplex (EBS):** Resulting from mutations in the genes for **keratin 5 or keratin 14**.

○ **Epidermolytic Hyperkeratosis (EHS):** Resulting from mutations in **keratin 1 or keratin 10**.

IV. Biological Regulation and Production

○ **Keratinocytes:** These are the specialized epithelial cells responsible for the synthesis of keratin.

○ **Retinoid Control:** The expression of keratin genes is regulated by **Vitamin A (retinoic acid)**. Retinoic acid binds to nuclear receptors that interact with chromatin to control the production of specific proteins, including keratin, in epithelial tissues.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Collagen

🏠 is the most abundant protein in the human body and serves as a fundamental structural component of the extracellular matrix (ECM). It is characterized by its unique three-dimensional architecture and specialized mechanical properties tailored to various tissues.

I. Molecular Structure

Triple Helix: A typical collagen molecule is a long, rigid structure composed of **three polypeptide chains** (referred to as "**-chains**") wound around each other in a rope-like **triple helix**.

Stabilization: These chains are held together by extensive **interchain hydrogen bonds**.

Amino Acid Sequence: Each -chain is approximately **1,000 amino acids long**. They feature a repeating sequence of **–Gly–X–Y–**, where glycine (the smallest amino acid) occupies every third position to fit into the restricted space where the three chains meet. Proline and hydroxyproline are also abundant, facilitating the "kinks" necessary for the helical conformation.

II. Classification and Types

The collagen superfamily includes **more than 25 types**, categorized based on their subunit composition and biological function.

Subunit Composition:

Type I Collagen: The most common form (90% of all collagen), typically containing **two chains and one chain** ().

Type II Collagen: Found primarily in cartilage, it consists of **three identical chains** ().

Functional Groups:

Fibril-forming collagens (Types I, II, III): These have a rope-like structure and aggregate into long fibrils with characteristic banding patterns.

Network-forming collagens (Types IV, VII): These form a **three-dimensional mesh or sheet** rather than distinct fibrils. For example, Type IV is a major component of **basement membranes**, providing support and acting as a semipermeable filtration barrier.

Fibril-associated collagens (Types IX, XII): These bind to the surface of collagen fibrils, linking them to each other and other ECM components.

III. Tissue-Specific Organization

Collagen organization is dictated by its structural role in a particular organ:

Tendons: Bundled in tight, parallel fibers to provide **great tensile strength**.

Bone: Arranged as fibers at an angle to each other to **resist mechanical shear** from any direction.

Vitreous Humor and ECM: Exists as a **dispersed gel** to provide support.

Cornea: Stacked specifically to transmit light with minimal scattering.

IV. Clinical Significance

Proper collagen function depends on **post-translational modifications**, such as the hydroxylation of proline and lysine residues, which requires **Vitamin C** as a cofactor.

Scurvy: A deficiency in Vitamin C results in unstable collagen helices and decreased tensile strength, leading to skin bruising and capillary fragility.

Ehlers-Danlos Syndrome (EDS): A group of inherited disorders resulting from defects in fibrillar collagen (often Type III), characterized by fragile, stretchy skin and loose joints.

Osteogenesis Imperfecta (OI): Also known as **brittle bone syndrome**, this is typically

caused by mutations in Type I collagen genes, leading to bones that easily fracture.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Elastin

The Elastic Fibrous Protein

✓ Unlike collagen, which is characterized by toughness and high tensile strength, **elastin** is a connective tissue protein with unique **rubber-like properties**.

▽ **Location and Function:** Elastin is found in the lungs, the walls of large arteries (like the **aorta**), and elastic ligaments. It allows these tissues to stretch to several times their normal length and recoil back to their original shape when the force is relaxed.

▽ **Structure and Precursor:** It is synthesized from a linear polypeptide precursor called **tropoelastin** (approximately 700 amino acids long). Tropoelastin is primarily composed of **small, nonpolar amino acids** such as **glycine, alanine, and valine**.

▽ **Assembly:** Upon secretion into the extracellular space, tropoelastin interacts with glycoprotein microfibrils, specifically **fibrillin**, which acts as a scaffold for elastin deposition.

▽ **Cross-linking and Desmosine:** Elastin forms an interconnected, rubbery network through the formation of **desmosine cross-links**. These are produced when three allysyl side chains (oxidized lysine) and one unaltered lysyl side chain from neighboring polypeptides condense together.

▽ **Amino Acid Composition:** Elastin is rich in proline and lysine but, unlike collagen, contains very little hydroxyproline or hydroxylysine.

II. Clinical Comment: Emphysema and -Antitrypsin

☉ **Emphysema** is a chronic obstructive lung disease that can result from a deficiency in **-antitrypsin (-AT)**.

Role of -AT: This protein, primarily synthesized in the liver, is the most important inhibitor of **neutrophil elastase**—a powerful protease that degrades elastin in alveolar walls.

☉ **Smokers and -AT:** Cigarette smoking causes the oxidation and inactivation of a specific methionine residue in -AT required for binding to elastase. Consequently, smokers with a genetic -AT deficiency experience significantly accelerated lung destruction.

☉ **Liver Involvement:** Genetic defects can also cause the mutated -AT protein to polymerize and remain trapped in the liver, potentially leading to **cirrhosis**.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Globular Proteins: Characteristics and Solubility

▽ **Globular proteins** are characterized by their compact, roughly **spherical shape**, which results from complex folding of the polypeptide chain.

☉ **Structural Organization:** In an aqueous environment, these proteins fold to bury **hydrophobic amino acid side chains in the interior** (the hydrophobic core) while keeping **hydrophilic (polar or charged) side chains on the outside surface**.

🔗 **Solubility:** Because of their polar surfaces, globular proteins are usually **very soluble** in aqueous solutions.

🔗 **Biological Roles and Examples:** This class includes nearly all **enzymes**, transport proteins like **hemoglobin** and **myoglobin**, and various **hormones**.

🔗 **Subclasses:** Based on your query and previous discussion, major subclasses include **albumins** (water-soluble and heat-coagulable), **globulins** (soluble in dilute salt solutions), and **histones** (basic, water-soluble proteins that bind to DNA).

[24/09/2026 10:46 AM] Imaanaa Harmee: Nucleic Acids

[24/09/2026 10:46 AM] Imaanaa Harmee:

Nucleic acids

▽ are macromolecular polymers required for the **storage and expression of genetic information**.

▽ There are two chemically distinct types of nucleic acids found in living systems:

✓ **deoxyribonucleic acid (DNA)**

✓ **ribonucleic acid (RNA)**

The sequence of nitrogenous bases in DNA acts as **the genetic master plan**, while RNA molecules serve as the **"working copies" of this plan** to express proteins and perform specialized cellular functions.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Who invented DNA fingerprinting?

◇ **Sir Alec Jeffreys**, a British geneticist at the **University of Leicester**, developed DNA fingerprinting in **1984**.

◇ He was **knighted in 1994**.

◇ The technique was initially based on variation in **minisatellite (VNTR) DNA sequences**.

First criminal case

◆ **Colin Pitchfork** became the **first person convicted using DNA fingerprinting evidence**.

◆ He was arrested in **1987** after DNA profiling linked him to the rape and murder of **two girls in Leicestershire, England**.

◆ DNA profiling also helped **exonerate an innocent suspect**, Richard Buckland, who had initially been suspected.

3. Elizabeth Hurley case

◆ In **2002**, DNA testing established that **Steve Bing was the biological father of Elizabeth Hurley's son, Damian**.

◆ She is owner of "Elizabeth Pink" Lipstick company

O.J. Simpson case

✂ O.J. Simpson was charged in **1994** with the murders of **Nicole Brown Simpson and Ronald Goldman**. (American footballer & actor) was **exonerated a double murder charge of his ex-wife & her boyfriend**.

✂ DNA evidence was a major part of the prosecution's case.

✂ However, Simpson was **acquitted in the criminal trial in 1994**.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Myostatin & the “Double-Muscle” Phenotype

▽ **Myostatin** is a member of the **TGF- β (transforming growth factor- β) family** that acts as a **negative regulator of skeletal muscle growth**.

Myostatin = “muscle growth brake” ●

★ Normally:

Myostatin $\uparrow$ → inhibits muscle development → limits muscle mass

● When myostatin function is **lost or defective**:

Myostatin $\downarrow$ /absent → inhibition removed → excessive skeletal muscle growth → “double-muscle” phenotype

Generally

MYO-STATIN = MYO-STOP ●

Myostatin stops muscle growth.

No myostatin → muscle grows excessively.

[24/09/2026 10:46 AM] Imaanaa Harmee:

Overview & Biological Functions

▽ **Nucleic acids** are macromolecular polymers required for the **storage and expression of genetic information**¹. The flow of genetic information inside a cell is described by the "**central dogma**" of molecular biology: **DNA is replicated** to copy and pass genetic information to daughter cells, **transcribed into RNA** (the "working copies" of the genes), and subsequently **translated into functional proteins**.

Monomeric Building Blocks: Nucleotides & Nucleosides

The monomeric units of nucleic acids are **nucleotides**.

Nucleotide Structure: Each nucleotide is composed of a **nitrogenous base**, a **pentose monosaccharide**, and **one, two, or three phosphate groups** linked to the sugar.

✓ **Nucleoside vs. Nucleotide:** Adding a pentose sugar to a nitrogenous base forms a **nucleoside** (via an *N*-glycosidic bond to the anomeric carbon of the sugar). Esterifying one or

more phosphate groups to the 5'-hydroxyl group of the pentose sugar converts the nucleoside into a **nucleotide**.

✓ **Acidity and Charge:** The phosphate groups carry **negative charges** at physiological pH, which causes DNA and RNA to act as acids, giving them the name "nucleic acids".

✓ **Alternative Monomeric Roles:** Beyond constructing polymers, monomeric nucleotides serve several crucial cellular functions:

- ▽ They serve as the "**energy currency**" of the cell (principally **ATP**).
- ▽ They act as **activated carriers** of metabolic intermediates in the synthesis of carbohydrates and lipids (e.g. **UDP-glucose** and **CDP-choline**).
- ▽ They are structural components of **essential coenzymes** (e.g., **Coenzyme A**, **FAD**, **NAD⁺**, and **NADP⁺**).
- ▽ They act as **second messengers** in signal transduction pathways (e.g., **cAMP** and **cGMP**).
- ▽ They serve as **important regulatory compounds** that inhibit or activate key metabolic enzymes.

The Bases & Pentose Sugars

The heterocyclic bases and sugars are structurally distinct between DNA and RNA:

✓ **The Nitrogenous Bases:** These belong to two families of compounds:

✓ **Purines:** Both DNA and RNA contain **adenine (A)** and **guanine (G)**.

✓ **Pyrimidines:** Both DNA and RNA contain **cytosine (C)**. However, DNA contains **thymine (T)**, whereas RNA contains **uracil (U)**. Thymine differs structurally from uracil only by possessing a **methyl group** at position 5 of its ring.

✓ **Modified/Unusual Bases:** Modified bases (e.g., methylated, acetylated, or reduced bases like dihydrouracil) occur occasionally in tRNA and some viral DNA. These modifications protect them from nuclease degradation or aid in enzyme recognition.

The Pentose Sugars:

⚗ DNA contains **2'-deoxyribose**, which lacks a hydroxyl group at the carbon 2' position.

⚗ RNA contains **ribose**, which possesses a polar **hydroxyl group (-OH)** at the carbon 2' position.

DNA Structure & Physical Properties

⚗ **The Phosphodiester Backbone:** DNA is a linear polymer of deoxyribonucleoside monophosphates linked covalently by **3'→5' phosphodiester bonds**. This backbone has a distinct **polarity** with a **5'-end** (carrying a free phosphate) and a **3'-end** (carrying a free hydroxyl). By convention, sequence readouts are always written from **5' to 3'**.

⚗ **The Double Helix:** DNA exists as a double-stranded molecule where two complementary strands wind around each other in a **right-handed double helix**. The two strands run in opposite, **antiparallel directions** (one is 5'→3' and the other is 3'→5').

☞ **Base Pairing (Chargaff's Rule):** The hydrophobic bases project inward into the core of the helix, where a purine on one strand hydrogen-bonds with a pyrimidine on the opposite strand²¹more_horiz.

Adenine (A) pairs with Thymine (T) via two hydrogen bonds.

Guanine (G) pairs with Cytosine (C) via three hydrogen bonds.

This strict pairing dictates **Chargaff's Rule:** in any double-stranded DNA sample, A = T, G = C, and the total amount of purines equals the total amount of pyrimidines.

[24/09/2026 10:46 AM] Imaanaa Harmee: ☞ **Denaturation & Melting (T_m):** Disrupting the hydrogen bonds (via heat or alkaline pH) causes the complementary strands to separate, or **denature**. The temperature at which half of the helical structure is lost is defined as the **melting temperature (T_m)**. Molecules with higher G-C content have a higher T_m because G-C pairs are held by three hydrogen bonds²⁵²⁷. Single-stranded DNA (ssDNA) has a **higher relative absorbance of UV light at 260 nm** than double-stranded DNA (dsDNA).

Structural Conformations:

▽ **B-DNA:** The standard right-handed helix with 10 residues per 360° turn²⁸. Chromosomal DNA is primarily B-DNA.

▽ **A-DNA:** A right-handed helix with 11 base pairs per turn, produced by moderately dehydrating the B-form. This conformation is typical in DNA-RNA hybrids or double-stranded RNA regions.

▽ **Z-DNA:** A left-handed helix with 12 base pairs per turn, characterized by a "zigzagging" backbone²⁸. It occurs naturally in regions of alternating purines and pyrimidines (like poly-GC) and may play a role in regulating gene expression.

Subcellular Organization & Packaging

DNA is packaged differently depending on the cell type:

✳ **Eukaryotic DNA:** Nuclear genomic DNA is composed of long, **linear molecules**. To fit inside the nucleus, the DNA wraps around small, positively charged basic proteins called **histones** (rich in arginine and lysine) to form structural units called **nucleosomes** ("beads on a string"). This nucleoprotein complex is called **chromatin**.

✳ **Extranuclear Eukaryotic DNA:** Eukaryotes contain small, closed, **circular double-stranded DNA** molecules in their **mitochondria** and chloroplasts. Mitochondrial DNA and its protein-synthesizing systems closely resemble those of bacteria.

✳ **Prokaryotic DNA:** Prokaryotic cells contain a single, double-stranded, supercoiled **circular chromosome** condensed with non-histone proteins to form a **nucleoid**. They also frequently contain small, circular, extrachromosomal DNA elements called **plasmids**.

RNA Structure & Functional Types

Unlike DNA, RNA is typically **single-stranded**, much smaller, and lacks a continuous helical structure. However, single-stranded RNA molecules fold back on themselves to create extensive, irregular, complementary loop structures that are vital for binding proteins and catalyzing reactions.

There are three major classes of RNA that participate directly in protein synthesis:

♦**Ribosomal RNA (rRNA):** rRNA is the most abundant type of RNA in the cell. It associates with proteins to form **ribosomes**, the physical machinery of translation. In eukaryotic cells, rRNA is transcribed and assembled in the **nucleolus**. Some rRNAs act as true catalysts—for example, **peptidyl transferase** catalyzes peptide bond formation during translation. Because of this catalytic activity, they are termed **ribozymes**.

♦**Transfer RNA (tRNA):** tRNA is the smallest class (4S) and serves as an "**adaptor**" during protein synthesis. Each tRNA binds a specific amino acid covalently at its **3'-CCA terminal adenosine**. During translation, a unique trinucleotide sequence on the tRNA called an **anticodon** complementary base-pairs with a corresponding codon on the mRNA, ensuring the correct amino acid is incorporated into the growing polypeptide chain. tRNAs contain a very high percentage of unusual bases (like dihydrouracil and pseudouracil) and assume a classic **cloverleaf secondary structure**.

♦**Messenger RNA (mRNA):** mRNA carries the genetic sequence information from nuclear DNA to the cytoplasm to serve as the template for translation.

♦**Eukaryotic mRNA is monocistronic** (carries information for only one gene) and is post-transcriptionally modified with a protective **5'-cap** (7-methylguanosine attached "backward" via an unusual 5'→5' triphosphate linkage) and a **3'-poly-A tail** to stabilize the molecule. Eukaryotic pre-mRNA contains noncoding **introns** that must be spliced out by **spliceosomes** (composed of snRNAs and proteins) to join coding **exons** together.

[24/09/2026 10:46 AM] Imaanaa Harmee: ♦**Prokaryotic mRNA is often polycistronic** (carries information from multiple genes)⁴³. Because prokaryotes lack a nucleus, translation occurs simultaneously on the mRNA as it is being transcribed (coupled transcription-translation). Prokaryotic mRNA is not capped, lacks poly-A tails, and does not contain introns.

The Chemical Difference in Alkali Sensitivity

✓ The presence of the **2'-hydroxyl group (-OH)** on ribose makes **RNA highly susceptible to alkaline hydrolysis**¹⁴²⁶. Under basic conditions, the 2'-hydroxyl group on the ribose ring loses its proton, enabling the negatively charged oxygen to perform a nucleophilic attack on the adjacent phosphate, breaking the phosphodiester linkage.

✓ In contrast, **DNA is completely resistant to alkaline cleavage** because it contains 2'-deoxyribose, which lacks this 2'-hydroxyl group. While alkali causes the double strands of DNA to separate (denature), its covalent phosphodiester bonds remain intact. This chemical difference is exploited in laboratory techniques to selectively degrade RNA and isolate intact DNA.

[24/09/2026 10:52 AM] Imaanaa Harmee:

Nitrogenous bases

▽ also referred to as heterocyclic bases

▽ are nitrogen-containing heterocyclic ring structures that serve as the fundamental monomeric components of nucleosides, nucleotides, and nucleic acids.

▽ At physiological pH, the nitrogen atoms in these rings are weak bases that can form hydrogen bonds, accept or donate electrons, and frequently carry a positive charge.

These bases are structurally categorized into two major families: **purines** and **pyrimidines**.

1. Purines (Double-Ring Structures)

♦**Structure:** Composed of a fused double-ring system containing a six-membered and a five-membered heterocyclic ring.

♦**Members:** **Adenine (A)** and **Guanine (G)** are the principal purines found in both DNA and RNA.

♦**Synthesis Precursors:** The atoms making up the purine ring are derived from amino acids (aspartic acid, glycine, and glutamine), carbon dioxide CO₂, and N¹⁰ {N⁵-methyltetrahydrofolate}.

♦**Metabolic Fate:** Unlike pyrimidines, the purine ring cannot be cleaved by human cells. Instead, purines are metabolized and oxidized to **uric acid**. Overproduction or underexcretion of uric acid leads to hyperuricemia and the deposition of sodium urate crystals in joints, resulting in **gout**.

2. Pyrimidines (Single-Ring Structures)

♦**Structure:** Composed of a single six-membered heterocyclic ring.

♦**Members:**

Cytosine (C): Found in both DNA and RNA.

Thymine (T): Found exclusively in DNA.

Uracil (U): Found exclusively in RNA.

♦**Chemical Difference (T vs. U):** Thymine and uracil are structurally identical except that thymine possesses a **methyl group** at position 5 of its ring.

♦**Synthesis Precursors:** The pyrimidine ring is synthesized from glutamine, CO₂, and aspartic acid before being attached to an activated ribose (PRPP).

♦**Metabolic Fate:** Pyrimidine rings can be fully opened and degraded in human cells to highly soluble, non-toxic products: beta-**alanine** and beta-**aminoisobutyrate**, alongside ammonia and CO₂. They do not generate uric acid.

3. Unusual and Modified Bases

♦Modified bases are produced via post-transcriptional enzymatic alterations of the standard bases.

Common examples include **dihydrouracil (D)** (where a double bond of the uracil ring is reduced), **pseudouridine (psi)** (where uracil is attached to ribose by a carbon-carbon bond

rather than a nitrogen–carbon bond), **ribothymidine (T)**, and various methylated, acetylated, or glycosylated bases.

◆ These modifications are highly concentrated in **transfer RNA (tRNA)**. They help the cell recognize tRNA molecules or protect them from enzymatic nuclease degradation.

◆ **Hypoxanthine (I)** is an unusual purine base generated during the degradation of adenine and guanine. Its corresponding nucleoside is **inosine**, which is found in the anticodon loop of certain tRNAs where it facilitates wobble base-pairing with multiple codons on mRNA.

[24/09/2026 10:52 AM] Imaanaa Harmee:

Minor (Unusual) Bases in Nucleic Acids

▽ **Minor or modified bases** are structurally altered forms of the standard purines (adenine, guanine) and pyrimidines (cytosine, thymine, uracil). They are produced through post-transcriptional or post-replicative enzymatic modifications, such as **methylation, acetylation, glycosylation, or reduction**.

Distribution across Taxa & Organisms:

Viruses & Bacteriophages: Some viral DNA and bacteriophage nucleic acids naturally incorporate modified bases. For example, certain bacteriophages contain **5-hydroxymethylcytosine** in place of standard cytosine to protect their genomes from host bacterial restriction endonucleases.

Bacteria & Human DNA: 5-Methylcytosine is a major methylated base found in both bacterial and eukaryotic genomes. In bacteria, cytosine and adenine methylation marks self-DNA to prevent cleavage by native restriction enzymes. In humans, 5-methylcytosine occurs in **CpG islands** near gene promoters, where hypermethylation serves as an epigenetic mechanism to silence gene expression.

Transfer RNA (tRNA): tRNA molecules contain an exceptionally high concentration of unusual bases (10% to 20% of total nucleotides). Examples include **dihydrouracil (D)**, **pseudouridine (Psi)**, **ribothymidine (rT)**, and methylated derivatives.

Mammalian Messenger RNA (mRNA): Eukaryotic pre-mRNA undergoes extensive nuclear modifications, including the addition of a **N⁷-methylguanosine cap** at its 5'-end attached via an unusual **5'→5'** triphosphate linkage. Other modified purines found in mammalian mRNA and RNA species include **N⁶-methyladenine** and **N⁶,N⁶-dimethyladenine**.

Biological Significance:

- ✂ Modified bases aid in sequence recognition by specific regulatory proteins.
- ✂ They protect DNA and RNA molecules from hydrolytic degradation by nucleases.
- ✂ In tRNA, unusual bases facilitate proper 3D tertiary folding and expand codon-anticodon recognition via wobble base-pairing.

2. Plant-Derived Purine Derivatives (Methylxanthines)

In plants, methylated derivatives of **xanthine** (an intermediate in the purine degradation pathway) accumulate as biologically active alkaloids known as **methylxanthines**:

Theophylline (1,3-Dimethylxanthine): Naturally occurring purine derivative found in **tea leaves**.

Theobromine (3,7-Dimethylxanthine): Major purine derivative found in **cocoa beans/chocolate**.

Caffeine (1,3,7-Trimethylxanthine): Widely distributed purine alkaloid found in **coffee beans, tea, and guarana**.

Biochemical Action & Clinical Significance:

▽ These plant methylxanthines act as potent competitive inhibitors of **cAMP phosphodiesterase**, the enzyme responsible for converting cyclic AMP (cAMP) into inactive 5'-AMP.

▽ By blocking phosphodiesterase, methylxanthines keep intracellular **cAMP levels elevated** for extended periods.

This prolongs the intracellular downstream signaling effects of hormones like epinephrine and glucagon, stimulating **glycogenolysis** in the liver and **lipolysis** in adipose tissue.

3. Phosphate Groups (Derived from Phosphoric Acid)

Inorganic **phosphate (Pi)**, derived from phosphoric acid, is the third fundamental constituent of every nucleotide monomer:

Chemical Linkage: A phosphate group is attached by an ester linkage to the 5'-hydroxyl group of the pentose sugar (ribose or 2-deoxyribose) to form a nucleoside 5'-monophosphate (such as AMP or dAMP).

Acidic Property & Charge: Each attached phosphate group possesses ionizable hydroxyl groups that dissociate protons at physiological pH. This coats the outer surface of DNA and RNA double helices with **strong negative charges**, giving these polymers their acidic character and the name "**nucleic acids**".

High-Energy Bonds: Adding a second or third phosphate group generates nucleoside diphosphates (NDPs) and triphosphates (NTPs). The terminal phosphate groups are connected by **phosphoanhydride ("high-energy") bonds**.

[24/09/2026 10:52 AM] Imaanaa Harmee: **Energy Currency & Biosynthesis**: The cleavage of these high-energy phosphate bonds (such as in ATP, GTP, UTP, and CTP) yields substantial free energy to drive anabolic pathways, nucleic acid polymerization, active ion transport, and mechanical cellular work

Theophylline:-1,3-dimethyl Xanthine in tea.

Theobromine:-3,7-dimethyl Xanthine in cocoa.

Caffeine:-1,3,7-trimethyl Xanthine in coffee.

[24/09/2026 10:52 AM] Imaanaa Harmee:

Nucleoside

▽ is a structural unit composed of a nitrogenous base (a purine or pyrimidine) covalently joined to a pentose sugar (ribose or 2-deoxyribose).

Type of Bond: The base is linked to the anomeric carbon (carbon 1') of the pentose sugar through a β -N-glycosidic bond.

Difference from Nucleotides: A nucleoside contains **no phosphate groups**. The addition of one, two, or three phosphate groups to the 5'-hydroxyl group of the nucleoside sugar converts it into a **nucleotide** (a nucleoside mono-, di-, or triphosphate).

Classification: Ribonucleosides vs. Deoxyribonucleosides

Nucleosides are classified into two groups based on the identity of their pentose sugar:

☞ **Ribonucleosides:** Contain the sugar **ribose** (which possesses a hydroxyl group on the 2'-carbon).

☞ **Deoxyribonucleosides:** Contain the sugar **2-deoxyribose** (which lacks a hydroxyl group at the 2'-carbon position).

Nomenclature & Examples

The naming of a nucleoside depends on whether the nitrogenous base is a purine or a pyrimidine:

Purine Nucleosides (end in -osine):

Adenine + Ribose = **Adenosine**

Guanine + Ribose = **Guanosine**

Hypoxanthine + Ribose = **Inosine**

Pyrimidine Nucleosides (end in -idine):

Cytosine + Ribose = **Cytidine**

Uracil + Ribose = **Uridine**

Thymine + Deoxyribose = **Deoxythymidine** (commonly referred to simply as **Thymidine**)

📌 *Note:* For deoxyribonucleosides, the prefix "**deoxy-**" is added before the purine or pyrimidine nucleoside name (e.g., **deoxyadenosine**, **deoxyguanosine**, **deoxycytidine**).

The Prime (') Numbering System

◇ To distinguish the atoms of the sugar from those of the nitrogenous ring, the carbons in

the pentose sugar are numbered with **primes (')**: 1', 2', 3', 4', 5'.

◇ The nitrogenous base ring atoms are numbered using standard integers (1, 2, 3...). Therefore, referencing the **5'-carbon** specifically designates the 5'-position of the pentose sugar, where phosphate groups attach.

[24/09/2026 10:53 AM] Mike: [nucleotide_structure_and_function.pdf]

[24/09/2026 10:54 AM] Imaanaa Harmee: ☞ Analogues of nucleobases, nucleosides & nucleotides

★ important for Cancer & HIV/AIDS Rx, gout treatment?

☞ Basis of Chemotherapy: altering a base or sugar moiety → induce toxic effect (Inhibition of NA synthesis).

Cytarabine (Arabinosyl-cytosine) & Vidarabine (arabinosyladenine):

☞ These nucleosides contain “Arabinose” in place of ribose.

☞ to Rx cancer & some viral infections

☞ Allopurinol: Rx of Gout

[24/09/2026 10:54 AM] Imaanaa Harmee:

Classification of Nucleic Acids

▽ Nucleic acids are classified primarily by their **sugar composition and base identity** into two major chemical classes—**Deoxyribonucleic Acid (DNA)** and **Ribonucleic Acid (RNA)**—and further categorized by their **subcellular location, structural conformation, and specialized cellular functions**.

1. Primary Chemical Classification: DNA vs. RNA

★ Deoxyribonucleic Acid (DNA):

Pentose Sugar: Contains **2'-deoxyribose** (lacking a hydroxyl group at carbon 2').

Nitrogenous Bases: Uses the purines **Adenine (A)** and **Guanine (G)**, and the pyrimidines **Cytosine (C)** and **Thymine (T)**.

Chemical Stability: Resistant to alkaline hydrolysis because it lacks a 2'-hydroxyl group.

Primary Role: Serves as the stable, permanent repository of genetic information.

★ Ribonucleic Acid (RNA):

Pentose Sugar: Contains **D-ribose** (possessing a free, polar hydroxyl group at carbon 2').

Nitrogenous Bases: Uses **Adenine (A)**, **Guanine (G)**, **Cytosine (C)**, and **Uracil (U)** instead of thymine (uracil lacks the 5-methyl group found in thymine).

Chemical Stability: Highly susceptible to **alkaline cleavage** because basic conditions cause the 2'-hydroxyl group to lose its proton and initiate a nucleophilic attack on the phosphodiester backbone.

Primary Role: Serves as the transient "working copies" of DNA to direct protein synthesis and perform structural, catalytic, or regulatory duties.

2. Classification of DNA (by Location, Host & Conformation)

A. Subcellular & Genomic Distribution

✧ **Eukaryotic Nuclear DNA:** Long, linear, double-stranded molecules associated with basic **histone proteins** (H1, H2A, H2B, H3, H4) to form **chromatin** packaged into chromosomes.

✧ **Mitochondrial DNA (mtDNA):** Closed, circular, double-stranded DNA located inside the mitochondrial matrix (<20,000 bp in humans); it replicates autonomously and closely resembles bacterial DNA systems.

✧ **Prokaryotic Chromosomal DNA:** A single, circular, supercoiled double-stranded chromosome condensed with non-histone proteins in a non-membrane-bound region called the **nucleoid**.

✧ **Plasmids:** Small, circular, extrachromosomal dsDNA molecules found in bacteria and yeast that replicate autonomously and often carry antibiotic resistance genes.

✧ **Viral Genomes:** Viruses contain either a DNA or RNA genome (never both), which can be single-stranded or double-stranded, linear or circular.

B. Structural Conformations of the Double Helix

✧ **B-DNA:** The classical right-handed double helix described by Watson and Crick (10 to 10.4 bp per turn), representing the predominant form of chromosomal DNA in vivo.

✧ **A-DNA:** A more compact right-handed helix (11 bp per turn) produced by dehydrating B-DNA; it is typical of **DNA-RNA hybrids** or double-stranded RNA regions.

✧ **Z-DNA:** A left-handed helix (12 bp per turn) with a "zigzagging" phosphate backbone, occurring transiently in alternating purine-pyrimidine sequences (e.g., poly-GC) and linked to gene transcription initiation.

3. Classification of RNA (by Biological Function)

A. Major Coding & Translational RNAs

✧ **Messenger RNA (mRNA) [~5% of total cellular RNA]:**
Carries genetic coding information from nuclear DNA to cytosolic ribosomes for translation.

✧ **Eukaryotic mRNA: Monocistronic** (codes for a single protein) and undergoes extensive post-transcriptional processing: a **5'-7-methylguanosine cap** (attached via an unusual 5'→5' linkage) and a **3'-poly-A tail** are added, and noncoding **introns** are spliced out.

✳ **Prokaryotic mRNA:** Often **polycistronic** (carries instructions for multiple proteins from an operon) and is translated simultaneously while being transcribed (coupled transcription-translation) without capping or tailing.

✳ **Ribosomal RNA (rRNA) [~80% of total cellular RNA]:**

Associates with proteins to form **ribosomes**, providing both structural framework and catalytic activity (e.g., **peptidyl transferase** ribozyme activity that forms peptide bonds).

Prokaryotic species: 23S, 16S, and 5S rRNAs.

Eukaryotic species: 28S, 18S, 5.8S, and 5S rRNAs (transcribed and assembled primarily in the nucleolus).

[24/09/2026 10:54 AM] Imaanaa Harmee: ✳ **Transfer RNA (tRNA) [~15% of total cellular RNA]:**

Smallest major class (~4S, ~80 nucleotides) acting as an "**adaptor**" during translation.

Features a characteristic **cloverleaf secondary structure**, a high proportion of **unusual/modified bases** (e.g., dihydrouracil, pseudouridine), an amino acid attachment site (**3'-CCA sequence**), and an **anticodon loop** that complementary base-pairs with mRNA codons.

B. Small & Regulatory Noncoding RNAs (ncRNAs)

✳ **Small Nuclear RNA (snRNA):** Complexes with proteins to form **snRNPs ("snurps")** inside the nucleus to mediate pre-mRNA splicing (removing introns and joining exons).

✳ **Small Nucleolar RNA (snoRNA):** Resides in the nucleolus to direct the cleavage, trimming, and modification of pre-rRNA precursors.

✳ **MicroRNAs (miRNAs) & Short Interfering RNAs (siRNAs):** Short (~22 bp) noncoding RNAs that associate with the **RISC complex** to mediate **RNA interference (RNAi)**, silencing specific target mRNAs by blocking translation or causing mRNA degradation.

✳ **RNA Primers:** Short RNA oligonucleotides synthesized by **primase** that provide the free 3'-OH group required by DNA polymerases to initiate DNA strand replication.

✳ **Ribozymes:** RNA molecules with enzymatic catalytic activity, such as self-splicing intron precursors or peptidyl transferase

[24/09/2026 10:57 AM] Imaanaa Harmee:

Enzymology

[24/09/2026 10:57 AM] Imaanaa Harmee:

📖 INTRODUCTION TO ENZYMES 🔪

Life depends on thousands of chemical reactions occurring every second.

These reactions include:

 Digestion of food

 Muscle contraction

 Nerve impulse transmission

Heart contraction

 DNA replication

 Energy production

If these reactions occurred naturally without assistance, they would proceed too slowly to sustain life.

Enzymes accelerate the rate of biochemical reactions without being consumed during the reaction.

 Definition of an Enzyme

An enzyme is a biological catalyst that increases the rate of a chemical reaction without being permanently changed or consumed during the reaction.

Unlike ordinary chemical catalysts, enzymes function under the mild conditions present inside living cells.

Enzymes make biochemical reactions proceed rapidly enough to support life.

 Why Are Enzymes Important?

Enzymes are essential because they regulate nearly every biochemical process occurring in the body.

Examples include:

 Digestion of carbohydrates, proteins, and fats.

Production of energy within cells.

 DNA and RNA synthesis.

 Hormone synthesis.

 Immune defense.

🧠 Neurotransmitter production.

💪 Muscle contraction.

♥ Maintenance of normal body functions.

Without enzymes,
life would not be possible because biochemical reactions would occur far too slowly.
[24/09/2026 10:57 AM] Imaanaa Harmee:

⚡ Enzymes as Biological Catalysts

A catalyst is a substance that increases the speed of a chemical reaction without being used up.

Enzymes perform exactly this function within living organisms

After catalyzing one reaction,
the enzyme remains unchanged and can participate in another reaction repeatedly.

🦋 How Do Enzymes Speed Up Reactions?

Enzymes lower the activation energy required for a chemical reaction to occur.

Because the activation energy becomes lower,
more molecules can react in a given period,
making the reaction proceed much faster.

Importantly,
the enzyme does not change:

☒ The overall energy released by the reaction.

☒ The final products of the reaction.

It only increases the reaction rate.

🦋 *What Is Activation Energy?*

Activation energy is the minimum amount of energy required to initiate a chemical reaction.

Every biochemical reaction must overcome this energy barrier before products can be formed.

Enzymes lower this energy barrier, allowing reactions to occur rapidly.

Relationship Between Enzymes and Activation Energy

Without enzymes:

The activation energy is high.

Only a few substrate molecules possess sufficient energy to react.

The reaction proceeds slowly.

With enzymes:

The activation energy decreases.

Many more substrate molecules can react.

The reaction becomes much faster.

This is the fundamental reason enzymes are called biological catalysts.

Enzymes Participate in Metabolism

Enzymes are involved in all metabolic pathways.

Metabolism consists of all chemical reactions occurring within living cells.

It can be divided into two major categories:

Anabolism

Anabolism refers to the synthesis of larger molecules from smaller molecules.

Examples include:

 Protein synthesis.

 Glycogen formation.

 DNA synthesis.

Catabolism

Catabolism refers to the breakdown of complex molecules into simpler molecules to release

energy.

Examples include:

 Glucose breakdown.

 Protein digestion.

 Fat metabolism.

 Enzyme–Substrate Relationship

Enzymes act on substances known as substrates.

The substrate binds to the enzyme.

The enzyme converts the substrate into products.

The enzyme is released unchanged and can catalyze another reaction.

 Distribution of Enzymes

Enzymes are present in:

 Brain

 Heart

Lungs

Blood

 Bone

 Skeletal muscle

 Every living cell

Different cells contain different enzymes depending on their specialized functions.

 Importance in Medicine

The lecture introduces enzymes as molecules that are also important in clinical diagnosis.

Measurement of enzyme activity in blood can help identify tissue injury and disease.

[24/09/2026 10:57 AM] Imaanaa Harmee:

🧠 Memory Trick

Remember the word ENZYME:

🔪 E = Efficient catalyst

🧬 N = Not consumed

⚡ Z = Zero change in equilibrium

⚙️ Y = Yields products faster

😊 M = Metabolic reactions

🔗 E = Essential for life

[24/09/2026 10:57 AM] Imaanaa Harmee:

✳️ Major Properties of Enzymes

⚡ High Efficiency

Enzymes are highly efficient catalysts.

This means they can accelerate biochemical reactions tremendously compared with reactions occurring without enzymes.

Because enzymes are not consumed during catalysis, the same enzyme molecule can participate repeatedly in many reaction cycles.

💡 Why Is High Efficiency Important?

High efficiency allows the body to:

Produce energy continuously.

🧠 Support brain function.

🔗 Sustain muscle contraction.

🧬 Carry out thousands of metabolic reactions simultaneously.

Without highly efficient enzymes, these reactions would proceed too slowly to maintain life.

High Specificity

One of the most important characteristics of enzymes is their high specificity.

Each enzyme recognizes and acts only on its appropriate substrate.

This specificity prevents unnecessary or harmful biochemical reactions.

What Is Enzyme Specificity?

Enzyme specificity means:

One enzyme usually catalyzes one particular reaction or acts on one particular substrate.

This precise recognition ensures that metabolic pathways remain organized and efficient.

Mild Reaction Conditions

Enzymes function under mild reaction conditions.

These conditions are compatible with living cells and include normal physiological environments.

Unlike many laboratory chemical reactions, enzymes do not require extreme temperatures or harsh chemicals to function.

Induced-Fit Model of Enzyme Specificity

According to this model:

The enzyme's active site is flexible rather than completely rigid.

When the correct substrate approaches,
the enzyme undergoes a slight conformational change.

This change improves the fit between the enzyme and substrate, allowing efficient catalysis.

Active Site

The active site is the region of the enzyme where the substrate binds and the chemical reaction occurs.

Only substrates with the correct structural features can bind effectively to the active site.

Enzyme–Substrate Complex

After the substrate binds to the active site,
an enzyme–substrate complex is formed.

Within this complex,
the enzyme facilitates conversion of the substrate into product(s).

After the reaction,
the products are released,
and the enzyme remains unchanged, ready to catalyze another reaction.
[24/09/2026 10:57 AM] Imaanaa Harmee:

The Active Site

The active site is the specific region of an enzyme where:

- ☒ The substrate binds.
- ☒ Catalysis takes place.
- ☒ Products are formed.

Every enzyme possesses a unique active site that determines which substrate can bind.

Substrate

A substrate is the molecule upon which an enzyme acts.

Each enzyme recognizes only its own appropriate substrate because of the specific structure of its active site.

This specificity prevents unwanted biochemical reactions.

Enzyme–Substrate Complex (ES Complex)

The enzyme–substrate complex is a temporary association between an enzyme and its substrate.

Formation of this complex allows the enzyme to:

-  Position the substrate correctly.

⚡ Lower activation energy.

⚙️ Increase reaction efficiency.

After catalysis,
the products leave the active site,
and the enzyme is free to catalyze another reaction.

🧬 General Sequence of Enzyme Action

According to the lecture concepts, enzyme action follows this sequence:

- 1 The substrate approaches the enzyme.
- 2 The substrate binds to the active site.
- 3 The enzyme–substrate complex forms.
- 4 The enzyme lowers the activation energy.
- 5 The substrate is converted into product(s).
- 6 Products are released.
- 7 The enzyme remains unchanged and can catalyze another reaction.

[24/09/2026 10:57 AM] Imaanaa Harmee:

ENZYME

NOMENCLATURE

most enzyme names end with the suffix "-ase."

The enzyme may be named:

☒ After its substrate

or

☒ After the reaction it catalyzes.

Examples include:

🔑 Glucosidase

🔑 Urease

🔑 Lactate dehydrogenase

 Adenylyl cyclase

some enzymes retain traditional (trivial) names, such as:

 Trypsin

 Pepsin 

SYSTEMATIC CLASSIFICATION

The systematic naming system classifies enzymes according to the type of reaction they catalyze, forming six enzyme classes. 

 CLASS 1:

OXIDOREDUCTASES

 Definition

oxidoreductases catalyze oxidation–reduction (redox) reactions. 

These reactions involve the transfer of electrons between molecules.

Characteristics

☒ Catalyze oxidation and reduction.

☒ Frequently require electron carriers.

☒ Participate in energy metabolism.

 Example

 NADH-dependent reactions.

 CLASS 2:

TRANSFERASES

 Definition

Transferases catalyze the transfer of functional groups from one molecule to another.

✿ Functional Groups Commonly Transferred

Examples include:

✎ Phosphate groups

✎ Amino groups

✎ Methyl groups

✎ Example

✎ Kinases

③ CLASS 3:

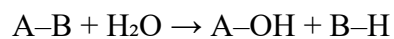
HYDROLASES 💧

📖 Definition

hydrolases catalyze hydrolytic cleavage reactions.

Water participates directly in the reaction.

General reaction given:



✎ Examples

✎ Esterases

✎ Glycosidases

✎ Peptidases

✎ Phosphatases

④

CLASS 4:

LYASES ✂

📖 Definition

lyases catalyze:

☑ Removal of a group.

☑ Addition of a group to a double bond.

They cleave without water, often producing a double bond.

🔗 Example

🔑 Decarboxylases

5

CLASS 5:

ISOMERASES 📄

📖 Definition

isomerases catalyze intramolecular rearrangements.

They rearrange atoms within the same molecule to form an isomer.

6 CLASS 6:

LIGASES 🔗

📖 Definition

ligases catalyze reactions in which two molecules are joined together.

These reactions require ATP-dependent bond formation (condensation).

🔗 Example

🔑 Pyruvate carboxylase

✳️ SUMMARY OF THE SIX CLASSES

1️⃣ Oxidoreductases → Oxidation–reduction reactions

2️⃣ Transferases → Transfer functional groups

3️⃣ Hydrolases → Hydrolytic cleavage

4️⃣ Lyases → Cleave or add groups without water

5 Isomerases → Intramolecular rearrangement

6 Ligases → Join two molecules using ATP

[24/09/2026 10:57 AM] Imaanaa Harmee:
ENZYME KINETICS

enzyme kinetics is the study of the rate (velocity) of enzyme-catalyzed reactions and the factors that influence this rate.

Studying enzyme kinetics helps us understand:

🧬 How enzymes work.

🔬 How enzymes are regulated.

💊 How drugs inhibit enzymes.

How diseases affect enzyme activity.

📌 ENZYME ACTIVITY

enzymes are not expressed by their concentration (such as mg or μg).
Instead, they are expressed by their activity.

📖 Definition

Enzyme activity is defined as:

The number of moles of substrate converted into product per unit time.

It can also be described as:

☑ Rate of product formation.

☑ Rate of substrate disappearance.

⚖ Units of Enzyme Activity

the SI unit is the Katal (kat).

📌 Definition of Katal

1 katal (1 kat) is the amount of enzyme that converts 1 mole of substrate into product in one second.

🌟 Enzyme Kinetics

enzymes are the most efficient catalysts known and do not change the equilibrium of the reaction they catalyze.

📖 Factors Affecting Enzyme Activity

enzyme activity is influenced by:

🔑 Substrate concentration

🔑 Temperature

⚖️ pH

🧬 Enzyme concentration

Activators

🚫 Inhibitors

🔑 Michaelis–Menten Theory

in 1913, Leonor Michaelis and Maud Menten proposed the Michaelis–Menten equation, based on the existence of an intermediate enzyme–substrate (ES) complex.

this model describes the relationship between:

📊 Reaction velocity (V)

and

🔑 Substrate concentration ([S]).

📖 Michaelis–Menten Equation

$$V = V_{\max} \times [S] / ([S] + K_m)$$

Where:

V = Initial reaction velocity

$V_{\max}$ = Maximum reaction velocity

$[S]$ = Substrate concentration

K_m = Michaelis constant

Enzyme–Substrate Complex

The lecture describes the reaction sequence as:



Where:

E = Enzyme

S = Substrate

ES = Enzyme–substrate complex

P = Product

This equation explains that enzyme and substrate first combine to form an intermediate complex before product formation.

Effect of Substrate Concentration

reaction velocity increases as substrate concentration increases because more enzyme active sites become occupied.

Initially,
many enzyme molecules are free.
As substrate concentration rises,
enzyme–substrate complexes become more common,
leading to faster product formation.

Initial Phase

At low substrate concentration,
most enzyme active sites remain free.
Therefore,
reaction velocity increases almost linearly with substrate concentration.

Saturation Phase

As substrate concentration continues to increase,
more active sites become occupied.

Eventually,
all active sites become filled.

At this stage,
further increases in substrate concentration do not increase reaction velocity.

Maximum Velocity (V_{max})

V_{max} is the maximum reaction velocity obtained when the enzyme becomes saturated with substrate.

At V_{max} ,
every enzyme active site is occupied.

No additional increase in reaction rate is possible by adding more substrate.

Michaelis Constant (K_m)

The lecture defines K_m as:

The substrate concentration at which the reaction velocity is one-half of V_{max} .

This is one of the most important definitions in enzyme kinetics.

Significance of K_m

K_m reflects the affinity between enzyme and substrate.

Low K_m

Low K_m indicates:

☒ Strong enzyme–substrate affinity.

Only a small amount of substrate is needed.

High K_m

High K_m indicates:

✗ Weak enzyme–substrate affinity.

A much larger amount of substrate is required.

📊 Order of Reaction

🔍 First-Order Reaction

When:

$$[S] \ll K_m$$

Reaction velocity is almost proportional to substrate concentration.
This is called first-order kinetics.

🔍 Half-Maximal Velocity

When:

$$[S] = K_m$$

Reaction velocity equals:
 $\frac{1}{2} V_{\max}$

This relationship is commonly tested in examinations.

🔍 Zero-Order Reaction

When:

$$[S] \gg K_m$$

The enzyme becomes saturated.

Reaction velocity reaches $V_{\max}$.

Adding more substrate no longer increases reaction velocity.

This is called zero-order kinetics.

🧠 Memory Trick

Remember the relationship:

🔗 "Low K_m = Loves substrate." 💖

Low K_m = High affinity

[24/09/2026 10:57 AM] Imaanaa Harmee:
High K_m = Low affinity

[24/09/2026 10:57 AM] Imaanaa Harmee:
THE LINEWEAVER–BURK PLOT 

the Michaelis–Menten curve is hyperbolic.

Because V_{max} is approached asymptotically, its exact value cannot easily be determined directly from the curve.

To overcome this limitation, Hans Lineweaver and Dean Burk developed the Lineweaver–Burk (double-reciprocal) plot.

 What is the Lineweaver–Burk Plot?

The Lineweaver–Burk plot is a straight-line graphical representation obtained by taking the reciprocals of the Michaelis–Menten equation.

Instead of plotting:
Reaction velocity (V)
against
Substrate concentration ($[S]$)

the graph plots:

$1/V$

against

$1/[S]$

This produces a straight line that allows easier determination of kinetic constants.

 Why is the Lineweaver–Burk Plot Important?

the Lineweaver–Burk plot is a better method for determining V_{max} and K_m than the substrate saturation curve.

It allows accurate determination of:

☒ Maximum velocity (V_{max})

☒ Michaelis constant (K_m)

Characteristics of the Lineweaver–Burk Plot

 The equation is linear in $1/V$ and $1/[S]$.

When these values are plotted:

☒ Slope = K_m / V_{max}

☒ Y-intercept = $1 / V_{max}$

☒ X-intercept = $-1 / K_m$

These relationships are among the most commonly tested concepts in enzyme kinetics.

 Memory Trick

Remember the three important values with:

 "SKY"

S = Slope = K_m / V_{max}

K = Negative K_m on
X-axis

Y = $1 / V_{max}$

Summary of Michaelis–Menten Kinetics

the Michaelis–Menten model describes many enzyme reactions under the following assumptions:

☒ The reaction has only one substrate.

☒ The substrate concentration is much higher than the enzyme concentration.

☒ Only the initial reaction velocity (V_o) is measured.

? Why Measure Only the Initial Velocity?

the initial reaction rate is measured because:

 At the beginning of the reaction,
only substrate is present.

✎ Enzyme active sites are free.

✎ Substrate is rapidly converted into product.

These conditions allow accurate measurement of enzyme activity.